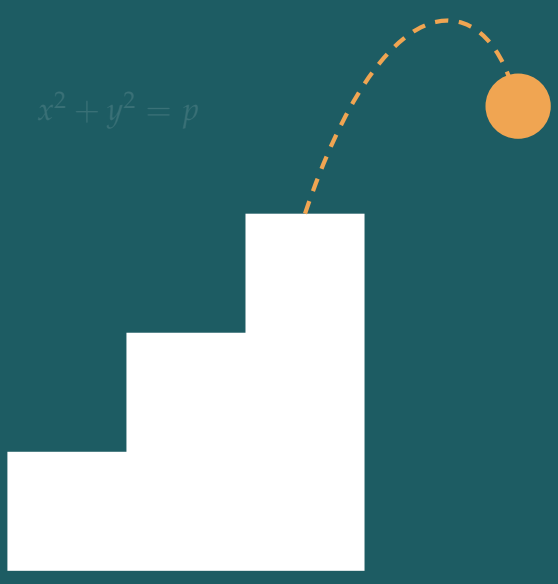
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP II AND III

2024 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP II AND III

2024 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2024 Worked Solutions

Worked solutions to all 24 questions of the 2024 STEP papers (STEP II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP II 2024, Question 1

Pure • Sums of Consecutive Integers

STEP II 2024, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Both sides are arithmetic series, so each is (number of terms) $\times$ (average term). Equating them gives one quadratic relation between n, k and c, and the rest of the question is number theory on that relation. Fixing k makes it easy to solve for c, and parity or remainders decide when c is a positive integer. Fixing $c = 1$ instead gives an equation of Pell type, and the map in (iv)(b) is the standard way to produce new solutions from old ones.

Solution

Part (i)

The $n + k$ integers from c upwards are $c, c+1, \ldots, c+n+k-1$. Their sum is

$$S_1 = (n+k)c + \frac{(n+k)(n+k-1)}{2}.$$

The next n integers are $c+n+k, \ldots, c+2n+k-1$, with sum

$$S_2 = n(c+n+k) + \frac{n(n-1)}{2}.$$

Doubling both sums and expanding,

$$\begin{aligned} 2S_1 &= 2nc + 2kc + n^2 + 2nk + k^2 - n - k, \\ 2S_2 &= 2nc + 2n^2 + 2nk + n^2 - n. \end{aligned}$$

Subtracting, $2S_1 - 2S_2 = 2kc + k^2 - k - 2n^2$. Hence

$$S_1 = S_2 \iff 2kc + k^2 - k = 2n^2 \iff 2n^2 + k = 2ck + k^2.$$

Every step is reversible (we only expanded, doubled and rearranged), so the equality of sums holds if and only if $2n^2 + k = 2ck + k^2$, as required.

Part (ii)

(a) With $k = 1$ the condition is $2n^2 + 1 = 2c + 1$, so $c = n^2$. This is a positive integer for every n, so *every* positive integer n works, with $c = n^2$. (For example, $n = 2$: $4 + 5 + 6 = 7 + 8$.)

(b) With $k = 2$ the condition is $2n^2 + 2 = 4c + 4$, that is,

$$n^2 = 2c + 1, \qquad c = \frac{n^2 - 1}{2}.$$

For c to be an integer, n^2 must be odd, so n must be odd. For c to be positive we need $n^2 > 1$, which rules out $n = 1$. Conversely every odd $n \geq 3$ gives a positive integer c. So the possible values are

$$n \in \{3, 5, 7, \dots\}, \qquad c = \tfrac{1}{2}(n^2 - 1).$$

The case $n = 3$, $c = 4$ is the equality $4 + 5 + 6 + 7 + 8 = 9 + 10 + 11$.

Part (iii)

With $k = 4$ the condition is $2n^2 + 4 = 8c + 16$, that is,

$$n^2 = 4c + 6 = 4(c + 1) + 2.$$

So n^2 would leave remainder 2 on division by 4. But a square leaves remainder 0 or 1: if $n = 2m$ then $n^2 = 4m^2$, and if $n = 2m + 1$ then $n^2 = 4(m^2 + m) + 1$. So there are no solutions.

Part (iv)

With $c = 1$ the condition becomes $2n^2 + k = 2k + k^2$, that is,

$$2n^2 = k(k + 1). \qquad (*)$$

In other words $n^2 = \frac{1}{2}k(k + 1)$: we want triangular numbers that are also perfect squares.

(a) Trying small k: $k = 1$ gives $2n^2 = 2$, so $n = 1$; then $k = 2, \dots, 7$ give $k(k + 1)/2 = 3, 6, 10, 15, 21, 28$, none of them squares; and $k = 8$ gives $2n^2 = 72$, so $n = 6$. So two possibilities are

$$k = 1,\ n = 1 \qquad \text{and} \qquad k = 8,\ n = 6.$$

(Check the second: $1 + 2 + \cdots + 14 = 105$ and $15 + 16 + \cdots + 20 = 105$.)

(b) Suppose $2N^2 = K^2 + K$. Put $N' = 3N + 2K + 1$ and $K' = 4N + 3K + 1$. Then

$$\begin{aligned} 2N'^2 &= 2(9N^2 + 4K^2 + 1 + 12NK + 6N + 4K) \\ &= 18N^2 + 8K^2 + 24NK + 12N + 8K + 2, \\ K'^2 + K' &= (16N^2 + 9K^2 + 1 + 24NK + 8N + 6K) + (4N + 3K + 1) \\ &= 16N^2 + 9K^2 + 24NK + 12N + 9K + 2. \end{aligned}$$

Subtracting,

$$2N'^2 - (K'^2 + K') = 2N^2 - K^2 - K = 0,$$

using the assumption. So $2N'^2 = K'(K'+1)$. Since N and K are positive integers, so are N' and K', and by $(*)$ and part (i) the value N' is a possible value of n, with corresponding $k = K'$.

> **Ask yourself** Where do these strange formulas come from?

Multiply $(*)$ by 4 and add 1: it becomes $(2k+1)^2 - 8n^2 = 1$. With $x = 2k+1$ and $y = n$ this is the Pell equation $x^2 - 8y^2 = 1$, whose solutions are generated by multiplying $x + y\sqrt{8}$ by the smallest solution $3 + \sqrt{8}$. Writing $(x + y\sqrt{8})(3 + \sqrt{8}) = (3x + 8y) + (x + 3y)\sqrt{8}$ and translating back to k and n gives exactly $N' = 3N + 2K + 1$ and $2K' + 1 = 3(2K+1) + 8N$, i.e. $K' = 4N + 3K + 1$.

(c) Applying the map to $(N, K) = (1, 1)$ just gives $N' = 6$, $K' = 8$, the second solution from (a). Applying it to $(N, K) = (6, 8)$:

$$N' = 18 + 16 + 1 = 35, \qquad K' = 24 + 24 + 1 = 49,$$

and applying it again to $(35, 49)$:

$$N' = 105 + 98 + 1 = 204, \qquad K' = 140 + 147 + 1 = 288.$$

So two further solutions are

$$k = 49,\ n = 35 \qquad \text{and} \qquad k = 288,\ n = 204.$$

Indeed $2 \cdot 35^2 = 2450 = 49 \cdot 50$ and $2 \cdot 204^2 = 83232 = 288 \cdot 289$.

Checks

- The opening example has $n = 3$, $k = 2$, $c = 4$: $2 \cdot 9 + 2 = 20 = 2 \cdot 4 \cdot 2 + 4$, so it satisfies the relation in (i), and fits (ii)(b) with $c = \frac{1}{2}(9 - 1)$.
- A computer search over all $n, k < 40$ and $c < 200$ confirmed that the two sums agree exactly when $2n^2 + k = 2ck + k^2$, and a search with $c = 1$ and $n, k < 400$ found only $(k, n) = (1, 1), (8, 6), (49, 35), (288, 204)$, matching (iv).
- In (iv)(b), note that the step from (N, K) to (N', K') preserves the quantity $2N^2 - K^2 - K$ for *any* N, K, which is a quick way to test your algebra with small numbers, say $N = K = 0$: then $N' = K' = 1$ and both values are 0.

Common slips

- Including $n = 1$ in (ii)(b): it gives $c = 0$, which is not a positive integer.
- In (i), miscounting where the second block starts: it begins at $c + n + k$, not $c + n + k + 1$.

- In (iv)(c), giving $k = 8$, $n = 6$ again, which is the first output of the map but was already found in (a).

Beyond STEP

The equation $x^2 - 8y^2 = 1$ is a Pell equation, and for any non-square d the solutions of $x^2 - dy^2 = 1$ all come from powers of a single fundamental solution, through multiplication of numbers $x + y\sqrt{d}$. This is the first glimpse of units in rings like $\mathbb{Z}[\sqrt{d}]$, a central topic in algebraic number theory. Davenport's *The Higher Arithmetic* treats Pell's equation through continued fractions.

WORKED SOLUTION

STEP II 2024, Question 2

Pure • Series and Integration

STEP II 2024, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Part (i) turns every integral $\int_0^1 x^m/(8+x^3)\,\mathrm{d}x$ into a series, so any series built from the same ingredients is a combination of these integrals. To evaluate a series in (ii) or (iii), split its bracket into terms of the form $1/(3k+m+1)$, recombine the matching integrands into one fraction, and integrate. The factorisation $8+x^3 = (x+2)(x^2-2x+4)$ is what makes the resulting integrals tractable.

Solution

Throughout, write

$$I_m = \int_0^1 \frac{x^m}{8+x^3}\,\mathrm{d}x.$$

Part (i)

Taking out the factor 8 and using $(1+u)^{-1} = \sum_{k\geq 0}(-1)^k u^k$ for $|u|<1$ with $u = x^3/8$,

$$(8+x^3)^{-1} = \frac{1}{8}\left(1+\frac{x^3}{8}\right)^{-1} = \sum_{k=0}^{\infty}\frac{(-1)^k x^{3k}}{8^{k+1}} = \sum_{k=0}^{\infty}\frac{(-1)^k}{2^{3(k+1)}}\,x^{3k}.$$

This is valid when $|x^3/8|<1$, that is, when $|x|<2$.

Now multiply by x^m and integrate term by term over $[0,1]$, which lies inside the interval of validity (and we are told not to worry about convergence):

$$I_m = \sum_{k=0}^{\infty}\frac{(-1)^k}{2^{3(k+1)}}\int_0^1 x^{3k+m}\,\mathrm{d}x = \sum_{k=0}^{\infty}\frac{(-1)^k}{2^{3(k+1)}}\cdot\frac{1}{3k+m+1},$$

since $\int_0^1 x^n\,\mathrm{d}x = 1/(n+1)$ with $n = 3k+m \geq 0$. That is the required result.

Part (ii)

The three denominators $3k+3$, $3k+2$ and $3k+1$ are $3k+m+1$ with $m = 2, 1, 0$ respectively. So by part (i) the series is

$$I_2 - 2I_1 + 4I_0 = \int_0^1 \frac{x^2 - 2x + 4}{8 + x^3}\,\mathrm{d}x.$$

? **Ask yourself** Why should $x^2 - 2x + 4$ cancel with anything?

Because $8 + x^3$ is a sum of two cubes: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ with $a = x$, $b = 2$ gives

$$8 + x^3 = (x+2)(x^2 - 2x + 4).$$

Hence the integrand simplifies to $\dfrac{1}{x+2}$, so the series equals $\displaystyle\int_0^1 \frac{1}{x+2}\,\mathrm{d}x$, as required. Finally,

$$\int_0^1 \frac{1}{x+2}\,\mathrm{d}x = \Big[\ln(x+2)\Big]_0^1 = \ln 3 - \ln 2 = \ln \tfrac{3}{2}.$$

Part (iii)

First split the bracket into partial fractions in k. Write $2k+1 = \alpha(3k+2) + \beta(3k+1)$. Putting $k = -\frac{1}{3}$ gives $\frac{1}{3} = \alpha$; putting $k = -\frac{2}{3}$ gives $-\frac{1}{3} = -\beta$, so $\beta = \frac{1}{3}$. Therefore

$$\frac{72(2k+1)}{(3k+1)(3k+2)} = \frac{24}{3k+1} + \frac{24}{3k+2},$$

and by part (i) (with $m = 0$ and $m = 1$) the series equals

$$24I_0 + 24I_1 = \int_0^1 \frac{24(1+x)}{(x+2)(x^2 - 2x + 4)}\,\mathrm{d}x.$$

Now split this integrand into partial fractions:

$$\frac{1+x}{(x+2)(x^2 - 2x + 4)} = \frac{A}{x+2} + \frac{Bx + C}{x^2 - 2x + 4},$$

so that

$$1 + x = A(x^2 - 2x + 4) + (Bx + C)(x+2).$$

Putting $x = -2$ gives $-1 = 12A$, so $A = -\frac{1}{12}$. Comparing coefficients of x^2 gives $0 = A + B$, so $B = \frac{1}{12}$. Comparing constant terms gives $1 = 4A + 2C$, so $C = \frac{2}{3}$. (The x coefficients check: $-2A + 2B + C = \frac{1}{6} + \frac{1}{6} + \frac{2}{3} = 1$.) Multiplying by 24,

$$\begin{aligned}\frac{24(1+x)}{8+x^3} &= -\frac{2}{x+2} + \frac{2x+16}{x^2 - 2x + 4} \\ &= -\frac{2}{x+2} + \frac{2x-2}{x^2 - 2x + 4} + \frac{18}{(x-1)^2 + 3},\end{aligned}$$

where we have split off the derivative $2x-2$ of the quadratic and completed the square in what remains.

We integrate each piece over $[0,1]$.

$$\int_0^1 \frac{-2}{x+2}\,\mathrm{d}x = -2\ln\tfrac{3}{2} = -2\ln 3 + 2\ln 2,$$
$$\int_0^1 \frac{2x-2}{x^2-2x+4}\,\mathrm{d}x = \Big[\ln(x^2-2x+4)\Big]_0^1 = \ln\tfrac{3}{4} = \ln 3 - 2\ln 2,$$
$$\int_0^1 \frac{18}{(x-1)^2+3}\,\mathrm{d}x = \frac{18}{\sqrt{3}}\Big[\arctan\frac{x-1}{\sqrt{3}}\Big]_0^1$$
$$= 6\sqrt{3}\Big(0 - \arctan\big(-\tfrac{1}{\sqrt{3}}\big)\Big) = 6\sqrt{3}\cdot\frac{\pi}{6} = \pi\sqrt{3}.$$

In the last line we used $\int \frac{\mathrm{d}u}{u^2+c^2} = \frac{1}{c}\arctan\frac{u}{c}$ with $u = x-1$, $c=\sqrt{3}$. Adding the three pieces, the $\ln 2$ terms cancel and

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^{3(k+1)}}\left(\frac{72(2k+1)}{(3k+1)(3k+2)}\right) = \pi\sqrt{3} - \ln 3,$$

so $a=3$ and $b=3$.

Checks

- The series converge very fast (each term is at most an eighth of the previous one), so a few terms are a good check. In (ii), the first three terms of the series give $0.41667 - 0.01198 + 0.00084 \approx 0.40553$, against $\ln\frac{3}{2} \approx 0.40547$.
- In (iii), $\pi\sqrt{3} - \ln 3 \approx 5.4414 - 1.0986 = 4.3428$; summing the first 200 terms of the series by computer agrees to twelve decimal places. Even the $k=0$ term alone gives $\frac{1}{8}\cdot 36 = 4.5$, the right size.

Common slips

- Writing the expansion with 8^{-k} instead of $8^{-(k+1)}$: the factor $\frac{1}{8}$ taken out at the start must not be lost.
- Matching $\frac{1}{3k+3}$ with I_3 rather than I_2: the denominator is $3k+m+1$, so m is one less than the constant.
- Forgetting that $\arctan\big(-\frac{1}{\sqrt{3}}\big)$ is negative, which gives the wrong sign for the $\pi\sqrt{3}$ term.

Beyond STEP

Swapping a sum and an integral, as in part (i), is legitimate here because the power series converges uniformly on $[0,1]$; when such swaps are allowed is a central theme

of first-year analysis. The same method sums $1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \cdots = \int_0^1 \frac{\mathrm{d}x}{1+x^3}$ exactly as in part (iii). Abbott's *Understanding Analysis* explains uniform convergence and term-by-term integration carefully.

WORKED SOLUTION

STEP II 2024, Question 3

Pure • Half-Angle Tangent Substitution

STEP II 2024, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Part (i) sets up a dictionary: the point at angle ϕ on the circle corresponds to the height $\tan\frac{1}{2}\phi$ where the line from N through it crosses the y-axis. Run it backwards too, so that every height q gives exactly one point on the circle. Then each function of q in the question becomes a trigonometric identity in $\frac{1}{2}\theta$, and we can read off what happens to the angle θ, and so to the point.

Solution

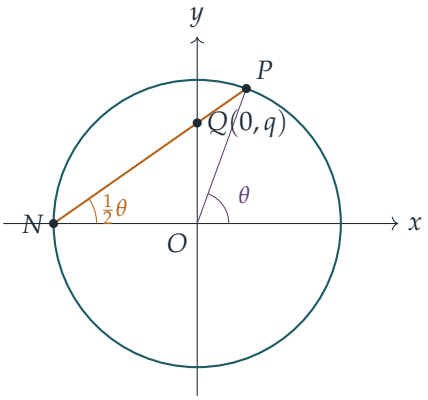
Part (i)

Since $P \neq N$, we have $\cos\theta \neq -1$, so the line NP is not vertical. Its gradient is

$$\frac{\sin\theta - 0}{\cos\theta - (-1)} = \frac{2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta}{2\cos^2\frac{1}{2}\theta} = \tan\tfrac{1}{2}\theta,$$

using the double-angle formulas $\sin\theta = 2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$ and $1+\cos\theta = 2\cos^2\frac{1}{2}\theta$ (and $\cos\frac{1}{2}\theta \neq 0$, since $|\frac{1}{2}\theta| < \frac{1}{2}\pi$). The line passes through $(-1,0)$, so it is $y = (x+1)\tan\frac{1}{2}\theta$, and putting $x = 0$ gives $q = \tan\frac{1}{2}\theta$, as required.

Geometrically, this is the angle at the circumference: the chord NP makes angle $\frac{1}{2}\theta$ with the x-axis at N because P subtends angle θ at the centre.



Going backwards: from a height to a point

We will keep needing the converse. Given any real q, the line through N and $(0,q)$ is $y = q(x+1)$. Substituting into $x^2+y^2=1$:

$$x^2 - 1 + q^2(x+1)^2 = 0 \iff (x+1)\left[(1+q^2)x - (1-q^2)\right] = 0.$$

The root $x=-1$ is N, so the other intersection is

$$\left(\frac{1-q^2}{1+q^2},\ \frac{2q}{1+q^2}\right).$$

If $q = \tan\frac{1}{2}\phi$ then by the t-formulas this point is $(\cos\phi, \sin\phi)$. Since $\phi \mapsto \tan\frac{1}{2}\phi$ is a bijection from $(-\pi,\pi)$ to $\mathbb{R}$, we get the following.

Dictionary. If $\tan\frac{1}{2}\phi = q$ for *any* angle ϕ (not necessarily in $(-\pi,\pi)$), then the line from N through $(0,q)$ meets the circle again at $(\cos\phi, \sin\phi)$.

This holds for every ϕ because replacing ϕ by $\phi \pm 2\pi$ changes neither $\tan\frac{1}{2}\phi$ (which has period π in $\frac{1}{2}\phi$) nor the point.

Part (ii)

(a) Since $\tan\frac{1}{4}\pi = 1$, the addition formula gives

$$\tan\left(\tfrac{1}{2}\theta + \tfrac{1}{4}\pi\right) = \frac{\tan\frac{1}{2}\theta + 1}{1 - \tan\frac{1}{2}\theta} = \frac{1+q}{1-q} = \mathrm{f}_1(q),$$

which is $\tan\frac{1}{2}\left(\theta + \frac{1}{2}\pi\right)$, as required. The condition $q \neq 1$ means $\theta \neq \frac{1}{2}\pi$, so the denominator is non-zero.

(b) By (a), the height of Q_1 is $\tan\frac{1}{2}\phi$ with $\phi = \theta + \frac{1}{2}\pi$. By the dictionary, $P_1 = \left(\cos(\theta+\frac{1}{2}\pi), \sin(\theta+\frac{1}{2}\pi)\right)$. So

P_1 is the image of P under an anticlockwise rotation about O through $\frac{1}{2}\pi$.

(The excluded value $q=1$ is $\theta = \frac{1}{2}\pi$, the one point whose image under this rotation is N itself.)

Part (iii)

(a) P_2 is the point at angle $\theta + \frac{1}{3}\pi$, so by part (i), applied with the angle reduced into $(-\pi, \pi)$ if necessary (which doesn't change tan of half the angle), and $\tan\frac{1}{6}\pi = \frac{1}{\sqrt{3}}$:

$$\mathrm{f}_2(q) = \tan\left(\tfrac{1}{2}\theta + \tfrac{1}{6}\pi\right) = \frac{q + \frac{1}{\sqrt{3}}}{1 - \frac{q}{\sqrt{3}}} = \frac{\sqrt{3}q + 1}{\sqrt{3} - q}.$$

The condition $q \neq \sqrt{3}$ is $\theta \neq \frac{2}{3}\pi$, which is exactly when P_2 would be N and the line NP_2 wouldn't exist.

(b) This time use $\tan\frac{1}{4}\pi = 1$ with a difference:

$$\mathrm{f}_3(q) = \frac{1-q}{1+q} = \frac{\tan\frac{1}{4}\pi - \tan\frac{1}{2}\theta}{1 + \tan\frac{1}{4}\pi\tan\frac{1}{2}\theta} = \tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}\theta\right) = \tan\tfrac{1}{2}\left(\tfrac{1}{2}\pi - \theta\right).$$

By the dictionary, P_3 is the point at angle $\frac{1}{2}\pi - \theta$, namely $(\sin\theta, \cos\theta)$. The coordinates of P are swapped, so

$$P_3 \text{ is the reflection of } P \text{ in the line } y = x.$$

(c)

Ask yourself What does f_2^{-1} do to the angle?

From (a), f_2 sends $\tan\frac{1}{2}\phi$ to $\tan\frac{1}{2}(\phi + \frac{1}{3}\pi)$ for every angle ϕ (the calculation never used $\phi = \theta$). Hence $\mathrm{f}_2\big(\tan\frac{1}{2}(\psi - \frac{1}{3}\pi)\big) = \tan\frac{1}{2}\psi$, and since f_2 is one-to-one (solving $s = \mathrm{f}_2(q)$ gives the unique $q = \frac{\sqrt{3}s-1}{s+\sqrt{3}}$, for $s \neq -\sqrt{3}$),

$$\mathrm{f}_2^{-1}\big(\tan\tfrac{1}{2}\psi\big) = \tan\tfrac{1}{2}\left(\psi - \tfrac{1}{3}\pi\right).$$

So f_2^{-1} rotates the angle back by $\frac{1}{3}\pi$.

Now follow the angle through the three steps. As $0 < q < 1$, we have $0 < \theta < \frac{1}{2}\pi$.

- f_2: angle $\theta + \frac{1}{3}\pi \in (\frac{1}{3}\pi, \frac{5}{6}\pi)$. Here $\mathrm{f}_2(q) > 0$, so f_3 can be applied.
- f_3: angle $\frac{1}{2}\pi - (\theta + \frac{1}{3}\pi) = \frac{1}{6}\pi - \theta \in (-\frac{1}{3}\pi, \frac{1}{6}\pi)$. This is not $-\frac{2}{3}\pi$, so the height is not $-\sqrt{3}$ and f_2^{-1} can be applied.
- f_2^{-1}: angle $\frac{1}{6}\pi - \theta - \frac{1}{3}\pi = -\theta - \frac{1}{6}\pi$.

So $\mathrm{f}_4(q) = \tan\frac{1}{2}\big(-\theta - \frac{1}{6}\pi\big)$, and by the dictionary P_4 is the point at angle $-\frac{1}{6}\pi - \theta$.

Reflection in the line through O at angle α sends the point at angle θ to the point at angle $2\alpha - \theta$ (the mirror line bisects the two directions). Here $2\alpha = -\frac{1}{6}\pi$, so

P_4 is the reflection of P in the line through O at angle $-\frac{1}{12}\pi$ to the positive x-axis,

that is, the line $y = -\tan\frac{1}{12}\pi\, x = -(2-\sqrt{3})\, x$.

This makes sense: rotating by $\frac{1}{3}\pi$, reflecting in $y = x$ and rotating back is a reflection in the mirror $y = x$ rotated back by $\frac{1}{3}\pi$, and $\frac{1}{4}\pi - \frac{1}{3}\pi = -\frac{1}{12}\pi$. Explicitly, with $t = \tan\frac{1}{12}\pi = 2-\sqrt{3}$,

$$\mathrm{f}_4(q) = -\tan\left(\tfrac{1}{2}\theta + \tfrac{1}{12}\pi\right) = -\frac{q+2-\sqrt{3}}{1-(2-\sqrt{3})q}.$$

Checks

- Take $\theta = \frac{1}{2}\pi$ in (iii)(b): $P = (0,1)$, $q = 1$, $\mathrm{f}_3(1) = 0$, so $Q_3 = O$ and $P_3 = (1,0)$, which is indeed the reflection of $(0,1)$ in $y = x$.
- For (c) with $q = \frac{1}{2}$: $\mathrm{f}_2(\frac{1}{2}) = \frac{\sqrt{3}+2}{2\sqrt{3}-1} \approx 1.5146$, then $\mathrm{f}_3 \approx -0.2046$, then $\mathrm{f}_2^{-1} \approx -0.8868$, matching the closed form $-\frac{\frac{1}{2}+2-\sqrt{3}}{1-\frac{1}{2}(2-\sqrt{3})} \approx -0.8868$.
- A computer check over 100 000 random points P confirmed every relationship above (rotation by $\frac{1}{2}\pi$, the formula for f_2, reflection in $y = x$, and reflection in the line at angle $-\frac{1}{12}\pi$) to within rounding error.

Common slips

- Stating that P_1 is at angle $\theta + \frac{1}{2}\pi$ without noticing that this angle may lie outside $(-\pi, \pi)$; it is harmless only because $\tan\frac{1}{2}\phi$ and the point both repeat when ϕ changes by 2π.
- In (c), applying the rotation for f_2^{-1} in the wrong direction (adding $\frac{1}{3}\pi$ again), which gives angle $\frac{1}{2}\pi - \theta$ and so wrongly makes P_4 the same as P_3.

Beyond STEP

The map from the point at angle θ to $q = \tan\frac{1}{2}\theta$ is *stereographic projection* of the circle onto a line, and the functions f_1, f_2, f_3 are Möbius transformations $q \mapsto \frac{aq+b}{cq+d}$. Every rotation and reflection of the circle becomes such a transformation of the line, and composing the symmetries corresponds to multiplying 2×2 matrices $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Needham's *Visual Complex Analysis* develops this picture beautifully, extending it to the Riemann sphere.

WORKED SOLUTION

STEP II 2024, Question 4

Pure • Vectors and Angle Bisectors

STEP II 2024, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Scale everything down to unit vectors. If $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are the unit vectors along OX and OY, then $\hat{\mathbf{x}} + \hat{\mathbf{y}}$ is the diagonal of a rhombus, and the diagonal of a rhombus bisects its angle. The given $\mathbf{b}$ is just a positive multiple of $\hat{\mathbf{x}} + \hat{\mathbf{y}}$, so each bisecting vector is a positive multiple of a sum of two unit vectors. That makes the dot products in part (ii) collapse to a single symmetric expression.

Solution

Throughout, write θ for the angle XOY. Since O, X, Y are not collinear, $0 < \theta < \pi$, so

$$-1 < \cos\theta < 1.$$

Both strict inequalities get used below.

Part (i)(a)

Let $\hat{\mathbf{x}} = \mathbf{x}/|\mathbf{x}|$ and $\hat{\mathbf{y}} = \mathbf{y}/|\mathbf{y}|$, so $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \cos\theta$. Then

$$\mathbf{b} = |\mathbf{x}|\mathbf{y} + |\mathbf{y}|\mathbf{x} = |\mathbf{x}||\mathbf{y}|\,(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

This is a combination of $\mathbf{x}$ and $\mathbf{y}$, so it lies in the plane XOY. It is non-zero, because $\hat{\mathbf{x}} + \hat{\mathbf{y}} = \mathbf{0}$ would mean $\hat{\mathbf{y}} = -\hat{\mathbf{x}}$, which would make O, X, Y collinear.

Now compare the angles. We have

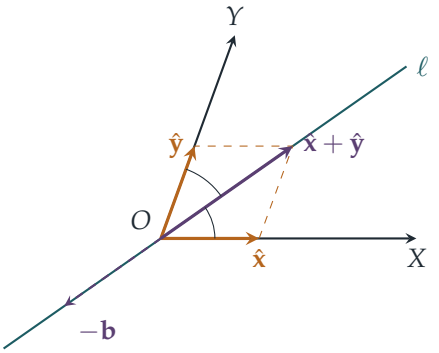
$$\mathbf{b} \cdot \hat{\mathbf{x}} = |\mathbf{x}||\mathbf{y}|\,(1 + \cos\theta) = \mathbf{b} \cdot \hat{\mathbf{y}},$$

so the angle α between $\mathbf{b}$ and $\overrightarrow{OX}$ and the angle β between $\mathbf{b}$ and $\overrightarrow{OY}$ satisfy

$$\cos\alpha = \frac{\mathbf{b} \cdot \hat{\mathbf{x}}}{|\mathbf{b}|} = \frac{|\mathbf{x}||\mathbf{y}|(1 + \cos\theta)}{|\mathbf{b}|} = \frac{\mathbf{b} \cdot \hat{\mathbf{y}}}{|\mathbf{b}|} = \cos\beta.$$

Both angles lie in $[0, \pi]$, where cosine is one-to-one, so $\alpha = \beta$. Also $1 + \cos\theta > 0$, so $\cos\alpha > 0$ and both angles are less than $90°$. Hence $\mathbf{b}$ is a bisecting vector for angle XOY.

Other bisecting vectors.



Since $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ have equal length, the parallelogram they span is a rhombus, and its diagonal $\hat{\mathbf{x}} + \hat{\mathbf{y}}$ splits angle XOY into two equal halves. In the plane XOY, the only directions making equal angles with OX and OY are the two directions along this diagonal line ℓ. If we rotate a vector off ℓ it moves closer to one arm and further from the other, so the angles become unequal. Along ℓ, the direction of $\mathbf{b}$ makes angle $\frac{1}{2}\theta < 90°$ with each arm. The opposite direction makes angle $180° - \frac{1}{2}\theta > 90°$ with each, so it is ruled out. So any bisecting vector points along ℓ in the same direction as $\mathbf{b}$, which means it is a positive multiple of $\mathbf{b}$.

The same argument in algebra: any vector in the plane is $\mathbf{v} = p\hat{\mathbf{x}} + q\hat{\mathbf{y}}$. Equal angles means $\mathbf{v} \cdot \hat{\mathbf{x}} = \mathbf{v} \cdot \hat{\mathbf{y}}$, and

$$\mathbf{v} \cdot \hat{\mathbf{x}} - \mathbf{v} \cdot \hat{\mathbf{y}} = (p + q\cos\theta) - (p\cos\theta + q) = (p - q)(1 - \cos\theta).$$

Since $\cos\theta \neq 1$, this forces $p = q$, so $\mathbf{v} = p(\hat{\mathbf{x}} + \hat{\mathbf{y}})$. The angles are acute only if $p > 0$.

Part (i)(b)

The point B lies on the line XY exactly when $\overrightarrow{OB} = (1 - t)\mathbf{x} + t\mathbf{y}$ for some real t. We need

$$\lambda|\mathbf{y}|\,\mathbf{x} + \lambda|\mathbf{x}|\,\mathbf{y} = (1 - t)\mathbf{x} + t\mathbf{y}.$$

Since $\mathbf{x}$ and $\mathbf{y}$ are not parallel, we may compare coefficients: $\lambda|\mathbf{y}| = 1 - t$ and $\lambda|\mathbf{x}| = t$. Adding these gives

$$\lambda = \frac{1}{|\mathbf{x}| + |\mathbf{y}|}, \qquad t = \frac{|\mathbf{x}|}{|\mathbf{x}| + |\mathbf{y}|}.$$

Then $\overrightarrow{XB} = t(\mathbf{y} - \mathbf{x})$ and $\overrightarrow{BY} = (1 - t)(\mathbf{y} - \mathbf{x})$ with $0 < t < 1$, so B lies between X and Y and

$$XB : BY = t : (1 - t) = |\mathbf{x}| : |\mathbf{y}| = OX : OY.$$

This is the angle bisector theorem: the bisector from O divides XY in the ratio of the adjacent sides.

Part (i)(c)

Here $\overrightarrow{OB}$ is a non-zero multiple of $\mathbf{b}$, so $OB \perp XY$ means $\mathbf{b}\cdot(\mathbf{y}-\mathbf{x}) = 0$. Using $\mathbf{b} = |\mathbf{x}||\mathbf{y}|(\hat{\mathbf{x}}+\hat{\mathbf{y}})$ and $\mathbf{y}-\mathbf{x} = |\mathbf{y}|\hat{\mathbf{y}} - |\mathbf{x}|\hat{\mathbf{x}}$,

$$\begin{aligned}\mathbf{b}\cdot(\mathbf{y}-\mathbf{x}) &= |\mathbf{x}||\mathbf{y}|\big(|\mathbf{y}|\cos\theta + |\mathbf{y}| - |\mathbf{x}| - |\mathbf{x}|\cos\theta\big)\\ &= |\mathbf{x}||\mathbf{y}|\,(|\mathbf{y}|-|\mathbf{x}|)(1+\cos\theta).\end{aligned}$$

Now $|\mathbf{x}|, |\mathbf{y}| > 0$ and $1+\cos\theta > 0$, so this is zero only if $|\mathbf{y}| = |\mathbf{x}|$. That is, $OX = OY$, and triangle XOY is isosceles.

Part (ii)

Let $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ be the unit vectors along $\overrightarrow{OP}$, $\overrightarrow{OQ}$, $\overrightarrow{OR}$. No three of the points are collinear, so in particular O, P, Q are not, and part (i)(a) applies to each angle. The chosen bisecting vectors must therefore be

$$\mathbf{u} = \alpha(\mathbf{p}+\mathbf{q}), \qquad \mathbf{v} = \beta(\mathbf{q}+\mathbf{r}), \qquad \mathbf{w} = \gamma(\mathbf{r}+\mathbf{p}),$$

for some $\alpha, \beta, \gamma > 0$, and all three are non-zero.

> **Ask yourself** The three bisecting vectors are arbitrary positive multiples. How can we say anything about all three angles at once?

The scale factors are positive, so they cannot change the sign of a dot product. Only the signs matter, since an angle between non-zero vectors is acute, right or obtuse according as their dot product is positive, zero or negative. Write $S = \mathbf{p}\cdot\mathbf{q} + \mathbf{q}\cdot\mathbf{r} + \mathbf{r}\cdot\mathbf{p}$. Using $\mathbf{q}\cdot\mathbf{q} = 1$,

$$\mathbf{u}\cdot\mathbf{v} = \alpha\beta\,(\mathbf{p}\cdot\mathbf{q} + \mathbf{p}\cdot\mathbf{r} + \mathbf{q}\cdot\mathbf{q} + \mathbf{q}\cdot\mathbf{r}) = \alpha\beta\,(1+S).$$

In exactly the same way, using $\mathbf{r}\cdot\mathbf{r} = 1$ and $\mathbf{p}\cdot\mathbf{p} = 1$,

$$\mathbf{v}\cdot\mathbf{w} = \beta\gamma\,(1+S), \qquad \mathbf{w}\cdot\mathbf{u} = \gamma\alpha\,(1+S).$$

Since $\alpha\beta$, $\beta\gamma$, $\gamma\alpha$ are all positive, the three dot products all have the sign of $1+S$. So the three angles are all acute if $1+S > 0$, all right angles if $1+S = 0$, and all obtuse if $1+S < 0$. ■

Checks

- Take $\mathbf{x} = (3,0,0)$ and $\mathbf{y} = (0,4,0)$. Then $\mathbf{b} = (12,12,0)$, which bisects the right angle. With $\lambda = \frac{1}{7}$ we get $B = (\frac{12}{7}, \frac{12}{7}, 0)$, which lies on $\frac{x}{3} + \frac{y}{4} = 1$, and $XB : BY = 3 : 4$.
- All three cases of part (ii) really occur. If P, Q, R are spread at 120° to each other in a plane, then $S = -\frac{3}{2}$ and the bisectors meet at 120° (obtuse). If $\mathbf{p}, \mathbf{q}, \mathbf{r}$ point to three vertices of a regular tetrahedron centred at O, each dot product is $-\frac{1}{3}$, so $S = -1$ and all three angles are right angles. Three nearly parallel vectors give acute angles.
- The results of both parts were also confirmed by computer on many thousands of random configurations.

Common slips

- Forgetting that $\cos\theta \neq \pm 1$ has to be used. The factor $1+\cos\theta$ in (i)(c) and the factor $1-\cos\theta$ in the uniqueness argument are non-zero only because the points are not collinear.
- In part (ii), treating the bisecting vectors as specific vectors such as $|\mathbf{x}|\mathbf{y}+|\mathbf{y}|\mathbf{x}$. The question lets them be chosen freely, so you have to quote part (i)(a) to say they are positive multiples of these.

Beyond STEP

Writing a vector as $|\mathbf{x}|\hat{\mathbf{x}}$ and working with unit vectors is a habit that pays off throughout linear algebra. The identity in (ii) is a small piece of the geometry of the *Gram matrix* of $\mathbf{p}, \mathbf{q}, \mathbf{r}$, whose entries are their pairwise dot products. The fact that $|\mathbf{p}+\mathbf{q}+\mathbf{r}|^2 = 3+2S \geq 0$ shows that $S \geq -\frac{3}{2}$, so the obtuse case has a limit. Strang's *Introduction to Linear Algebra* develops dot products, projections and Gram matrices in this spirit.

WORKED SOLUTION

STEP II 2024, Question 5

Pure • Ranges and Diophantine Equations

STEP II 2024, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

The range of a polynomial on $\mathbb{Z}$ doesn't change if we shift the variable by an integer, so we complete the square (or the cube) to put each function in its simplest shifted form. Asking whether two ranges meet then becomes asking whether an equation in two integer unknowns has a solution, and the trick is to rearrange that equation into the form *product of two integers = a small fixed number*. Factorising a small number has only a handful of possibilities, so we can check each one.

Solution

Part (i)

Completing the square,

$$\mathrm{f}_1(n) = (n+3)^2 + 2 = \mathrm{F}_1(n+3).$$

As n runs through $\mathbb{Z}$, so does $m = n+3$ (every integer m arises, from $n = m-3$). So the range of f_1 is $\{m^2+2 : m \in \mathbb{Z}\}$, which is exactly the range of F_1. That is what we needed.

Part (ii)

Similarly $\mathrm{g}_1(n) = (n-1)^2 + 4$, so by the same argument the range of g_1 is $\{k^2+4 : k \in \mathbb{Z}\}$. By part (i), a common value would need integers m, k with

$$m^2 + 2 = k^2 + 4 \quad \Longleftrightarrow \quad (m-k)(m+k) = 2.$$

Now $m-k$ and $m+k$ differ by $2k$, which is even, so they have the same parity. If both are odd, their product is odd, so it can't be 2. If both are even, their product is a multiple of 4, so again it can't be 2. Hence no such m, k exist, and the two ranges have empty intersection.

Part (iii)

Completing the square,

$$f_2(n) = (n-1)^2 - 7, \qquad g_2(n) = (n-2)^2 - 2,$$

so the ranges are $\{m^2 - 7\}$ and $\{k^2 - 2\}$ with m, k running over $\mathbb{Z}$. A common value needs

$$m^2 - 7 = k^2 - 2 \quad \iff \quad (m-k)(m+k) = 5.$$

Since 5 is prime, the pair $(m-k, m+k)$ is one of $(1,5)$, $(5,1)$, $(-1,-5)$, $(-5,-1)$. Adding the two factors gives $2m = \pm 6$, and subtracting gives $2k = \pm 4$, so $m^2 = 9$ and $k^2 = 4$ in every case. The common value is

$$m^2 - 7 = 9 - 7 = 2 = 4 - 2 = k^2 - 2.$$

So the only integer in both ranges is $\boxed{2}$. As a check, $f_2(4) = f_2(-2) = 2$ and $g_2(4) = g_2(0) = 2$.

Part (iv)

Completing the square in p,

$$p^2 + pq + q^2 = \left(p + \tfrac{1}{2}q\right)^2 + \tfrac{3}{4}q^2 \geq 0,$$

since both terms are squares times positive constants. (Equality holds only when $q = 0$ and $p = 0$.)

For the cubics, we compare f_3 with $(n-1)^3 = n^3 - 3n^2 + 3n - 1$:

$$f_3(n) = (n-1)^3 + 4n + 1 = (n-1)^3 + 4(n-1) + 5.$$

So, writing $m = n - 1$ (which again runs over all of $\mathbb{Z}$), the range of f_3 is $\{m^3 + 4m + 5\}$, and the range of g_3 is $\{k^3 + 4k - 6\}$. A common value needs integers m, k with

$$\begin{aligned} & m^3 + 4m + 5 = k^3 + 4k - 6 \\ \iff \quad & (k^3 - m^3) + 4(k-m) = 11 \\ \iff \quad & (k-m)\left(k^2 + km + m^2 + 4\right) = 11. \end{aligned}$$

Ask yourself How does the inequality at the start of this part help with this equation?

By the inequality with $p = k$, $q = m$, we have $k^2 + km + m^2 + 4 \geq 4$. In particular the second factor is a positive integer, so $k - m$ must be positive too. Since 11 is prime, there are only two options:

- $k - m = 11$ and $k^2 + km + m^2 + 4 = 1$. This is impossible, because the second factor is at least 4.

- $k - m = 1$ and $k^2 + km + m^2 + 4 = 11$.

In the second case, put $k = m + 1$:

$$\begin{aligned}(m+1)^2 + m(m+1) + m^2 &= 7\\ 3m^2 + 3m + 1 &= 7\\ m^2 + m - 2 = (m+2)(m-1) &= 0,\end{aligned}$$

so $m = 1$ (with $k = 2$) or $m = -2$ (with $k = -1$). The corresponding values are

$$m = 1:\ 1 + 4 + 5 = 10, \qquad m = -2:\ -8 - 8 + 5 = -11.$$

So the integers in both ranges are $\boxed{10 \text{ and } -11}$.

Returning to the original variables: $\mathrm{f}_3(2) = 8 - 12 + 14 = 10$ and $\mathrm{g}_3(2) = 8 + 8 - 6 = 10$; also $\mathrm{f}_3(-1) = -1 - 3 - 7 = -11$ and $\mathrm{g}_3(-1) = -1 - 4 - 6 = -11$. Interestingly, both common values come from the same input n.

Checks

- Substitute back, as we did above: f_2 and g_2 both take the value 2 at $n = 4$, and f_3, g_3 agree at $n = 2$ (value 10) and $n = -1$ (value -11).
- Listing small values is a quick sanity check for part (ii): f_1 gives $2, 3, 6, 11, 18, 27, \ldots$ and g_1 gives $4, 5, 8, 13, 20, 29, \ldots$. Neither list contains a number from the other.
- A computer search over all $|n| \leq 3000$ confirmed that the ranges in (i) agree, the ranges in (ii) are disjoint, and the only common values are 2 in (iii) and 10, -11 in (iv).

Common slips

- Finding a common value by solving $\mathrm{f}(n) = \mathrm{g}(n)$ for the *same* n. The two inputs can be different, so we need two independent unknowns. In (iii), $\mathrm{f}_2(n) = \mathrm{g}_2(n)$ happens to give the right answer, but the method is wrong.
- In (ii), arguing only that $(m - k)(m + k) = 2$ forces $\{m - k, m + k\} = \{1, 2\}$ and forgetting the negative factor pairs. The parity argument deals with every case at once.
- In (iv), forgetting to rule out the factorisation 11×1. It's the inequality that does this.

Beyond STEP

Equations like $m^2 - k^2 = 5$ and $k^3 + 4k = m^3 + 4m + 11$, where only integer solutions count, are *Diophantine equations*. Factorising one side and using the divisors of the other

is one of the most powerful elementary tools for solving them. The parity trick in (ii) generalises: an integer is a difference of two squares exactly when it is odd or a multiple of 4. For more, Davenport's *The Higher Arithmetic* is an excellent first-year introduction to number theory.

WORKED SOLUTION

STEP II 2024, Question 6

Pure • Binomial Series Coefficients

STEP II 2024, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every part runs on ratios of successive terms. A binomial coefficient $\binom{2n}{n}$ and a binomial-series coefficient are both products that grow by one factor at each step, so comparing the nth term with the $(n-1)$th gives a short recurrence. Two sequences that obey the same recurrence and start at the same value are equal. The final identity then comes from writing $(1-x)^{-3/2}$ as $(1-x)^{-1/2} \times (1-x)^{-1}$ and comparing coefficients, because multiplying by $(1-x)^{-1} = 1 + x + x^2 + \cdots$ turns a sequence of coefficients into its running totals.

Solution

Part (i)

Let $P(n)$ be the statement $T_n = \dfrac{1}{2^{2n}}\binom{2n}{n}$.

Base case. $P(0)$ holds, since $\frac{1}{2^0}\binom{0}{0} = 1 = T_0$.

Inductive step. Suppose $P(n-1)$ holds for some $n \geq 1$. Since $\binom{2n}{n} = \dfrac{(2n)!}{n!\,n!}$ and $\binom{2n-2}{n-1} = \dfrac{(2n-2)!}{(n-1)!\,(n-1)!}$, dividing gives

$$\binom{2n}{n} = \frac{(2n)(2n-1)}{n \cdot n}\binom{2n-2}{n-1} = \frac{2(2n-1)}{n}\binom{2n-2}{n-1}.$$

Then, using the recurrence and $P(n-1)$,

$$\begin{aligned} T_n = \frac{2n-1}{2n}\,T_{n-1} &= \frac{2n-1}{2n}\cdot\frac{1}{2^{2n-2}}\binom{2n-2}{n-1} \\ &= \frac{2n-1}{2n}\cdot\frac{1}{2^{2n-2}}\cdot\frac{n}{2(2n-1)}\binom{2n}{n} \\ &= \frac{1}{4\cdot 2^{2n-2}}\binom{2n}{n} = \frac{1}{2^{2n}}\binom{2n}{n}, \end{aligned}$$

which is $P(n)$. By induction, $P(n)$ holds for all $n = 0, 1, 2, \ldots$.

Part (ii)

By the binomial theorem for a general index, the coefficient of x^r in $(1-x)^{-1/2}$ is

$$a_r = \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{1}{2}-r+1\right)}{r!}(-1)^r = \frac{\frac{1}{2}\cdot\frac{3}{2}\cdots\frac{2r-1}{2}}{r!},$$

since each of the r factors in the numerator is negative and the $(-1)^r$ from $(-x)^r$ cancels the signs. Going from a_{r-1} to a_r adds one factor $\frac{2r-1}{2}$ on top and one factor r underneath, so for $r \geq 1$

$$\frac{a_r}{a_{r-1}} = \frac{(2r-1)/2}{r} = \frac{2r-1}{2r}, \qquad \text{that is,} \qquad a_r = \frac{2r-1}{2r}\,a_{r-1}.$$

Ask yourself Why does this recurrence immediately give $a_r = T_r$?

Both sequences start at 1 (because $a_0 = 1$, the constant term, and $T_0 = 1$) and obey the same rule for getting from term $r-1$ to term r. A one-line induction finishes it: if $a_{r-1} = T_{r-1}$, then $a_r = \frac{2r-1}{2r}a_{r-1} = \frac{2r-1}{2r}T_{r-1} = T_r$. So $a_r = T_r$ for all $r \geq 0$, and by (i),

$$a_r = \frac{1}{2^{2r}}\binom{2r}{r}.$$

Part (iii)

In the same way, the coefficient of x^r in $(1-x)^{-3/2}$ is

$$b_r = \frac{\frac{3}{2}\cdot\frac{5}{2}\cdots\frac{2r+1}{2}}{r!}.$$

The numerators of a_r and b_r are both products of r consecutive half-odd numbers. The product for b_r has lost the first factor $\frac{1}{2}$ of a_r and gained a factor $\frac{2r+1}{2}$ at the end. The $r!$ cancels, so

$$\frac{b_r}{a_r} = \frac{(2r+1)/2}{1/2} = 2r+1.$$

(For $r = 0$ both products are empty and $b_0 = a_0 = 1$, which agrees.) Hence

$$b_r = (2r+1)\,a_r = \frac{2r+1}{2^{2r}}\binom{2r}{r}, \qquad r = 0, 1, 2, \ldots.$$

Part (iv)

Since $(1-x)^{-3/2} = (1-x)^{-1/2}(1-x)^{-1}$ and $(1-x)^{-1} = \sum_{s\geq 0} x^s$,

$$\sum_{n=0}^{\infty} b_n x^n = \left(\sum_{r=0}^{\infty} a_r x^r\right)\left(\sum_{s=0}^{\infty} x^s\right).$$

On the right, a term x^n comes from choosing $a_r x^r$ from the first bracket and x^{n-r} from the second, for each $r = 0, 1, \ldots, n$. Each of these choices has coefficient $a_r \cdot 1$, so the coefficient of x^n is $a_0 + a_1 + \cdots + a_n$. Comparing coefficients of x^n on the two sides,

$$b_n = \sum_{r=0}^{n} a_r.$$

Substituting the formulae from (ii) and (iii) gives

$$\frac{2n+1}{2^{2n}}\binom{2n}{n} = \sum_{r=0}^{n} \frac{1}{2^{2r}}\binom{2r}{r},$$

as required. (The working holds for $n = 0$ too, where both sides equal 1.)

Checks

- Small cases: $a_0, a_1, a_2, a_3 = 1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}$, matching $\binom{2r}{r}/4^r = 1, \frac{2}{4}, \frac{6}{16}, \frac{20}{64}$.
- For $n = 3$, the left side of (iv) is $\frac{7\cdot 20}{64} = \frac{35}{16}$, and the right side is $1 + \frac{1}{2} + \frac{3}{8} + \frac{5}{16} = \frac{35}{16}$.
- Directly, $(1-x)^{-3/2} = 1 + \frac{3}{2}x + \frac{15}{8}x^2 + \cdots$, and $b_2 = 5 \cdot \frac{3}{8} = \frac{15}{8}$.
- A computer check with exact fractions confirmed (i)–(iv) for every $n \leq 29$. Summing the first 30 terms of each series at $x = 0.3$ gives 1.19523 and 1.70747, which agree with $(0.7)^{-1/2}$ and $(0.7)^{-3/2}$.

Common slips

- Sign errors in the binomial coefficients: $\binom{-1/2}{r}$ alternates in sign, but the factor $(-1)^r$ from $(-x)^r$ cancels this, so every a_r and b_r is positive.
- Forgetting the base case, or claiming $a_r = T_r$ from the recurrence alone: the starting values $a_0 = T_0 = 1$ must also agree.

Beyond STEP

Multiplying a power series by $(1-x)^{-1}$ to produce partial sums is a basic tool of *generating functions*, which turn questions about sequences into algebra with power series. The numbers $\binom{2n}{n}/4^n$ also appear in probability: this is the chance that a fair coin tossed $2n$ times shows exactly n heads, and Stirling's formula shows it behaves

like $1/\sqrt{\pi n}$ for large n. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit, and Grimmett and Welsh, *Probability: An Introduction*, develops generating functions properly.

WORKED SOLUTION

STEP II 2024, Question 7

Pure • Curve Sketching

STEP II 2024, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Each bracket is the equation of a circle, so C_1 is just two circles. For C_2, multiplying the brackets out collapses everything into $x^2 + y^2$ and x^2, so cutting with a horizontal line gives a quadratic in x^2. Counting intersections then comes down to counting the positive roots of that quadratic, which the discriminant, the sum and the product of the roots settle between them.

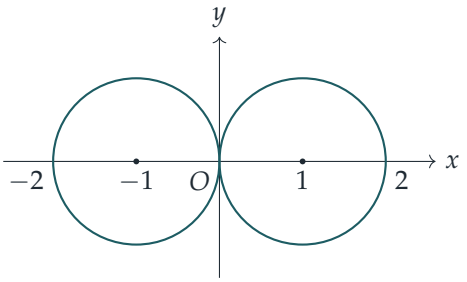
Solution

Throughout, write

$$A = y^2 + (x-1)^2 - 1 = x^2 + y^2 - 2x, \qquad B = y^2 + (x+1)^2 - 1 = x^2 + y^2 + 2x.$$

Part (i)

A product is zero exactly when one of its factors is zero, so C_1 is the union of the circles $A = 0$ and $B = 0$. These are the circles of radius 1 centred at $(1, 0)$ and $(-1, 0)$. Both pass through the origin, where they touch (their common tangent there is the y-axis). The highest points are $(\pm 1, 1)$ and the curve crosses the x-axis at $x = -2, 0, 2$.



Part (ii)(a)

Since A and B have the form $s-2x$ and $s+2x$ with $s=x^2+y^2$,

$$AB=(x^2+y^2)^2-4x^2.$$

On the line $y=k$ the equation $AB=\frac{1}{16}$ becomes

$$\begin{aligned}(x^2+k^2)^2-4x^2&=\tfrac{1}{16}\\ x^4+2k^2x^2+k^4-4x^2-\tfrac{1}{16}&=0\\ x^4+2(k^2-2)x^2+\left(k^4-\tfrac{1}{16}\right)&=0,\end{aligned}$$

as required.

Now put $u=x^2$, giving

$$u^2+2(k^2-2)u+\left(k^4-\tfrac{1}{16}\right)=0. \qquad (*)$$

Each root $u>0$ gives two intersection points $x=\pm\sqrt{u}$, a root $u=0$ gives one point ($x=0$), and a root $u<0$ gives none. So we need to know the signs of the roots of $(*)$. Three quantities tell us:

- the quarter-discriminant $(k^2-2)^2-\left(k^4-\frac{1}{16}\right)=\frac{65}{16}-4k^2$, which is positive, zero or negative as k is less than, equal to or greater than $\frac{\sqrt{65}}{8}$;
- the sum of the roots, $2(2-k^2)$;
- the product of the roots, $k^4-\frac{1}{16}$, which has the sign of $k-\frac{1}{2}$ for $k>0$.

Note that $\frac{1}{2}<\frac{\sqrt{65}}{8}\approx 1.008$, and that whenever the roots are real we have $k^2\leq\frac{65}{64}<2$, so their sum is positive.

> **Ask yourself** If the product of the roots is negative, do we need to check the discriminant?

No: a negative product forces the roots to be real (the discriminant b^2-4ac is then positive) and of opposite signs. Now we take the cases in turn.

- $0<k<\frac{1}{2}$: the product is negative, so there is exactly one positive root. **2 points.**
- $k=\frac{1}{2}$: $(*)$ becomes $u^2-\frac{7}{2}u=0$, so $u=0$ or $u=\frac{7}{2}$, giving $x=0$ and $x=\pm\sqrt{7/2}$. **3 points.**
- $\frac{1}{2}<k<\frac{\sqrt{65}}{8}$: the discriminant is positive, so there are two distinct real roots. Their product and sum are both positive, so both roots are positive. **4 points.**
- $k=\frac{\sqrt{65}}{8}$: there is a repeated root $u=2-k^2=\frac{63}{64}>0$, giving $x=\pm\frac{\sqrt{63}}{8}$. **2 points.**
- $k>\frac{\sqrt{65}}{8}$: no real roots. **No points.**

Part (ii)(b)

By (a), the line $y = k$ meets C_2 when $0 < k \leq \frac{\sqrt{65}}{8}$ and misses it for larger k. The equation of C_2 involves y only through y^2, so negative k gives nothing higher. The greatest y-coordinate on C_2 is therefore $\frac{\sqrt{65}}{8}$, attained at $x = \pm\frac{\sqrt{63}}{8}$.

On C_1 the highest points are the tops of the circles, $(\pm 1, 1)$. Since $\frac{\sqrt{63}}{8} < \frac{\sqrt{64}}{8} = 1$, the highest points of C_2 are **closer to the y-axis** than those of C_1 (though slightly higher, as $\frac{\sqrt{65}}{8} > 1$).

Part (ii)(c)

Suppose $A < 0$ and $B < 0$. From $A < 0$ we get $2x > x^2 + y^2 \geq 0$, so $x > 0$. From $B < 0$ we get $-2x > x^2 + y^2 \geq 0$, so $x < 0$. These contradict each other, so A and B cannot both be negative. (Geometrically, the two open discs don't overlap.)

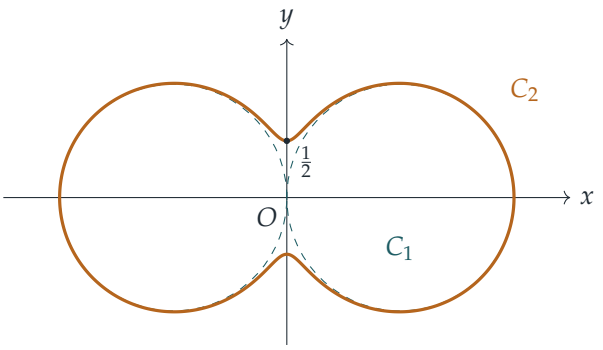
On C_2 we have $AB = \frac{1}{16} > 0$, so A and B are nonzero with the same sign. They can't both be negative, so both are positive. Now $A > 0$ says that $(x-1)^2 + y^2 > 1$, so the point lies strictly outside the circle centred at $(1, 0)$. Likewise $B > 0$ puts it strictly outside the circle centred at $(-1, 0)$. Hence every point of C_2 lies outside both circles: C_2 lies entirely outside C_1.

Part (ii)(d)

Useful features of C_2 for the sketch:

- It is symmetric in both axes, since only x^2 and y^2 appear.
- On $y = 0$, $(*)$ gives $u = 2 + \frac{\sqrt{65}}{4}$, so C_2 crosses the x-axis at $x = \pm\sqrt{2 + \frac{\sqrt{65}}{4}} \approx \pm 2.004$, just beyond ± 2.
- On $x = 0$, $k^4 = \frac{1}{16}$, so C_2 crosses the y-axis at $(0, \pm\frac{1}{2})$.
- Its highest and lowest points are $\left(\pm\frac{\sqrt{63}}{8}, \pm\frac{\sqrt{65}}{8}\right) \approx (\pm 0.99, \pm 1.01)$.

The counts in (a) show the shape. For $|y| < \frac{1}{2}$ there are just two points (the outer sides). For $\frac{1}{2} < |y| < \frac{\sqrt{65}}{8}$ there are four, so the curve dips down to $(0, \pm\frac{1}{2})$ between two humps. C_2 is therefore a single closed "peanut" enclosing both circles, pinched in at the y-axis.



(Writing C_2 in polar form gives $r^2 = 2\cos^2\theta + \sqrt{4\cos^4\theta + \frac{1}{16}}$, which is how the curve above was drawn.)

Checks

- At $k = \frac{\sqrt{65}}{8}$ the discriminant $\frac{65}{16} - 4k^2$ vanishes, matching the top of the curve. The repeated root $u = \frac{63}{64}$ gives $x^2 + y^2 = \frac{128}{64} = 2$ and $4x^2 = \frac{63}{16}$, and indeed $2^2 - \frac{63}{16} = \frac{1}{16}$.
- The point $(0, \frac{1}{2})$: $A = B = \frac{1}{4}$, so $AB = \frac{1}{16}$.
- Counting the crossings of $y = k$ with C_2 by computer for $k = 0.1$, 0.49, 0.51, 1.0, 1.007 and 1.01 gives 2, 2, 4, 4, 4 and 0 points respectively, and a numerical search finds the maximum height $1.00778 \approx \frac{\sqrt{65}}{8}$ at $x \approx 0.9922 \approx \frac{\sqrt{63}}{8}$.

Common slips

- Treating $u = 0$ as giving two points: $x = \pm\sqrt{0}$ is a single point, which is why $k = \frac{1}{2}$ gives 3 intersections, not 4.
- Forgetting the tangency case $k = \frac{\sqrt{65}}{8}$ in the count.
- In (c), concluding only "A and B have the same sign" without ruling out both negative, which is the whole point of the first half.

Beyond STEP

C_2 belongs to the family $(x^2 + y^2)^2 - 4x^2 = c$, the level curves of the function AB. As c increases the curve changes shape: two separate ovals inside the circles (for $c < 0$), then the figure-of-eight C_1 at $c = 0$, then a single peanut, whose dimples disappear once $c \geq 4$. The change of topology happens exactly at the level of the critical point of AB at the origin. Following how level curves change as they pass critical points is the starting idea of Morse theory, and the closely related Cassini ovals, with the lemniscate of Bernoulli as their figure-of-eight, behave in just the same way. Siklos, *Advanced*

Problems in Mathematics, has more STEP-style curve sketching of implicit curves like this.

WORKED SOLUTION

STEP II 2024, Question 8

Pure • Arithmetic-Geometric Mean

STEP II 2024, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.6, *The Arithmetic–Geometric Mean*: https://beyondstep.keithmcnulty.org/ch02/agm.html

The key idea

The two sequences squeeze together: the arithmetic means come down, the geometric means go up, and the gap at least halves each time, so both settle on a common limit M, the *arithmetic–geometric mean* of x_0 and y_0. In part (ii) the substitution turns the integral for the pair (p, q) into the integral for the pair of their means, so the integral never changes as we run the iteration. Letting $n \to \infty$ collapses it to an integral we can do.

Solution

Part (i)

We first show by induction that $0 < y_n < x_n$ for every n. It holds for $n = 0$. Suppose $0 < y_n < x_n$. Then $\sqrt{x_n} \neq \sqrt{y_n}$, so

$$0 < \left(\sqrt{x_n} - \sqrt{y_n}\right)^2 = x_n + y_n - 2\sqrt{x_n y_n},$$

which rearranges to $\frac{1}{2}(x_n + y_n) > \sqrt{x_n y_n}$, that is, $y_{n+1} < x_{n+1}$. Also $y_{n+1} = \sqrt{x_n y_n} > 0$. This completes the induction, so $y_{n+1} < x_{n+1}$ for all $n \geq 0$.

Using $y_n < x_n$:

$$x_{n+1} = \tfrac{1}{2}(x_n + y_n) < \tfrac{1}{2}(x_n + x_n) = x_n,$$
$$y_{n+1} = \sqrt{x_n y_n} > \sqrt{y_n y_n} = y_n,$$

where the second line uses $x_n > y_n > 0$ and the fact that the square root is increasing.

So (y_n) is increasing. It is also bounded above: $y_n < x_n < x_{n-1} < \cdots < x_0$, so $y_n < x_0$ for all n. By the theorem, $y_n \to M$ for some $M \leq x_0$. (Since the sequence increases from y_0, also $M \geq y_0 > 0$.)

For the gap, the left-hand inequality $x_{n+1} - y_{n+1} > 0$ is what we proved first. For the right-hand one, since $y_{n+1} > y_n$,

$$x_{n+1} - y_{n+1} < x_{n+1} - y_n = \tfrac{1}{2}(x_n + y_n) - y_n = \tfrac{1}{2}(x_n - y_n).$$

Applying this repeatedly,

$$0 < x_n - y_n < \frac{x_0 - y_0}{2^n} \to 0,$$

so $x_n - y_n \to 0$ by squeezing. Finally $x_n = y_n + (x_n - y_n)$ is the sum of a sequence tending to M and one tending to 0, so $x_n \to M$ as well.

Part (ii)

Write $a = \tfrac{1}{2}(p+q)$, $g = \sqrt{pq}$, and call the t-integral J.

The substitution. With $t = \tfrac{1}{2}\left(x - \frac{pq}{x}\right)$ for $x > 0$,

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{2}\left(1 + \frac{pq}{x^2}\right) = \frac{x^2 + pq}{2x^2} > 0,$$

so t increases strictly, from $-\infty$ as $x \to 0^+$ to $+\infty$ as $x \to \infty$: the substitution maps $(0, \infty)$ one-to-one onto the whole real line. Now we rewrite the two factors under the root.

$$pq + t^2 = \tfrac{1}{4}\left(x^2 + 2pq + \frac{p^2q^2}{x^2}\right) = \frac{1}{4}\left(x + \frac{pq}{x}\right)^2 = \left(\frac{x^2 + pq}{2x}\right)^2,$$

$$\begin{aligned}\tfrac{1}{4}(p+q)^2 + t^2 &= \tfrac{1}{4}\left(p^2 + 2pq + q^2 + x^2 - 2pq + \frac{p^2q^2}{x^2}\right) \\ &= \frac{x^4 + (p^2+q^2)x^2 + p^2q^2}{4x^2} = \frac{(x^2+p^2)(x^2+q^2)}{4x^2}.\end{aligned}$$

Since $x > 0$ and $x^2 + pq > 0$, the square root of the product is

$$\frac{x^2 + pq}{2x} \cdot \frac{\sqrt{(p^2+x^2)(q^2+x^2)}}{2x}.$$

Therefore

$$\begin{aligned}J &= \int_0^\infty \frac{x^2 + pq}{2x^2} \cdot \frac{4x^2}{(x^2+pq)\sqrt{(p^2+x^2)(q^2+x^2)}}\,\mathrm{d}x \\ &= 2\int_0^\infty \frac{\mathrm{d}x}{\sqrt{(p^2+x^2)(q^2+x^2)}} = 2\,\mathrm{I}(p,q).\end{aligned}$$

The same integral read directly. Since $\tfrac{1}{4}(p+q)^2 = a^2$ and $pq = g^2$, the integrand of J is $1/\sqrt{(a^2+t^2)(g^2+t^2)}$, which is even in t. So

$$J = 2\int_0^\infty \frac{\mathrm{d}t}{\sqrt{(a^2+t^2)(g^2+t^2)}} = 2\,\mathrm{I}(a,g),$$

which is legitimate because a, g are positive with $g < a$ (by the inequality of part (i), as $p \neq q$). Comparing the two expressions for J gives

$$\mathrm{I}(p,q) = \mathrm{I}\big(\mathrm{a}(p,q), \mathrm{g}(p,q)\big),$$

as required.

Evaluating $\mathrm{I}(x_0, y_0)$. Applying this with $(p,q) = (x_n, y_n)$, which is allowed since $0 < y_n < x_n$, gives $\mathrm{I}(x_n, y_n) = \mathrm{I}(x_{n+1}, y_{n+1})$, so

$$\mathrm{I}(x_0, y_0) = \mathrm{I}(x_n, y_n) \quad \text{for every } n.$$

Ask yourself We want to let $n \to \infty$ and say the integral tends to $\mathrm{I}(M, M)$. How can we justify moving the limit inside an integral over an infinite range?

We avoid any delicate limit argument by squeezing. For $0 < y_n < x_n$ and every x,

$$(y_n^2 + x^2)^2 \leq (x_n^2 + x^2)(y_n^2 + x^2) \leq (x_n^2 + x^2)^2,$$

so, taking square roots and reciprocals (which reverses the inequalities) and integrating,

$$\int_0^\infty \frac{\mathrm{d}x}{x_n^2 + x^2} \leq \mathrm{I}(x_n, y_n) \leq \int_0^\infty \frac{\mathrm{d}x}{y_n^2 + x^2}.$$

For $c > 0$, $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{c^2 + x^2} = \Big[\frac{1}{c} \arctan \frac{x}{c}\Big]_0^\infty = \frac{\pi}{2c}$, so

$$\frac{\pi}{2x_n} \leq \mathrm{I}(x_0, y_0) \leq \frac{\pi}{2y_n} \quad \text{for every } n.$$

The left-hand side is constant in n, and both bounds tend to $\pi/(2M)$ because $x_n \to M$ and $y_n \to M$ with $M > 0$. Hence

$$\mathrm{I}(x_0, y_0) = \frac{\pi}{2M}.$$

Checks

- Take $x_0 = 2$, $y_0 = 1$. The iteration gives $(1.5, 1.41421)$, then $(1.45711, 1.45648)$, and then both agree to five figures at $M \approx 1.45679$. So $\pi/(2M) \approx 1.07826$, and evaluating $\mathrm{I}(2,1)$ numerically gives 1.07826 too.
- If $p = q$ the identity is trivially true, since then $\mathrm{a}(p,p) = \mathrm{g}(p,p) = p$. The answer $\pi/(2M)$ is consistent with this, because $\mathrm{I}(M,M) = \pi/(2M)$.
- The convergence is extremely fast: after the first few steps the gap roughly squares each time, much better than the halving we proved.

Common slips

- Using $y_{n+1} < x_{n+1}$ in the induction without first establishing $y_n < x_n$ (and $y_n > 0$, so that the square roots make sense).
- Forgetting that the t-integral runs over the whole real line, and so equals *twice* $\mathrm{I}(a, g)$; the two factors of 2 must cancel.
- Writing $\mathrm{I}(x_n, y_n) \to \mathrm{I}(M, M)$ without any justification. The squeeze above gives an honest reason.

Beyond STEP

The common limit M is Gauss's arithmetic–geometric mean, and the integral I is a *complete elliptic integral*: Gauss found this identity in 1799, and it gives an astonishingly fast way to compute elliptic integrals, and even π itself (the Gauss–Legendre algorithm). The monotone convergence theorem quoted at the start, and the care needed when swapping limits and integrals, are central themes of a first course in analysis; Abbott's *Understanding Analysis* is an excellent place to meet them properly.

WORKED SOLUTION

STEP II 2024, Question 9

Mechanics • Projectile Motion

STEP II 2024, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

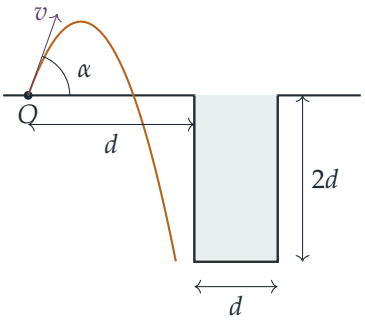
www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything follows from the trajectory equation $y = x\tan\alpha - \dfrac{gx^2}{2v^2}\sec^2\alpha$, written as a quadratic in $t = \tan\alpha$. Landing on the base cleanly means two things: the particle is still above ground level when it reaches the near edge, and it is already below the level of the base by the time it would reach the far side. Each condition is a quadratic inequality in t, and completing the square turns them into the forms we need. Part (ii) is then just a matter of showing the two allowed ranges of t overlap.

Solution

Model the particle as a point projectile under gravity alone, with no air resistance. Take the origin at the point of projection, x horizontally towards the trench and y vertically upwards. The near wall is then the line $x = d$, the far wall $x = 2d$, and the base $y = -2d$ for $d < x < 2d$.



With horizontal velocity $v\cos\alpha$ and vertical velocity $v\sin\alpha - gT$ at time T, eliminating T gives

$$y = x\tan\alpha - \frac{gx^2}{2v^2\cos^2\alpha} = xt - \frac{x^2(1+t^2)}{2\lambda d},$$

where $t = \tan\alpha$ and we have used $v^2 = \lambda dg$ and $\sec^2\alpha = 1 + t^2$.

Part (i)(a)

To avoid the ground before the trench and the near wall, the particle must still be strictly above ground level when $x = d$. (If it were at or below $y = 0$ there, it would already have struck the ground or would strike the near wall.) So

$$y(d) = dt - \frac{d(1+t^2)}{2\lambda} > 0 \iff 2\lambda t - 1 - t^2 > 0 \iff t^2 - 2\lambda t + \lambda^2 < \lambda^2 - 1,$$

multiplying by $2\lambda/d > 0$. That is,

$$(\tan\alpha - \lambda)^2 < \lambda^2 - 1,$$

as required. The left-hand side is a square, so $\lambda^2 - 1 > 0$; since $\lambda = v^2/(dg) > 0$, this gives $\lambda > 1$.

Part (i)(b)

Now consider the far wall. The path is a downward parabola which at $x = d$ is above the base level $-2d$. It reaches $y = -2d$ (the base) before $x = 2d$ exactly when it is already below that level at $x = 2d$, that is when $y(2d) < -2d$. If instead $y(2d) \geq -2d$, it arrives at $x = 2d$ without having reached the base, so it touches the far wall (or clears the trench altogether).

? **Ask yourself** Why is it enough to look at the single point $x = 2d$?

Because $y(x) + 2d$ is a quadratic in x with negative leading coefficient that is positive at $x = d$. It has exactly one root beyond $x = d$, and that root lies before $2d$ precisely when the quadratic is already negative at $x = 2d$.

So we need

$$y(2d) = 2dt - \frac{4d^2(1+t^2)}{2\lambda d} < -2d \iff \lambda t - (1+t^2) < -\lambda \iff t^2 - \lambda t + 1 - \lambda > 0,$$

multiplying by $\lambda/(2d) > 0$. Multiplying by 4 and completing the square,

$$4t^2 - 4\lambda t + \lambda^2 > \lambda^2 - 4 + 4\lambda, \qquad \text{i.e.} \qquad (2\tan\alpha - \lambda)^2 > \lambda^2 + 4(\lambda - 1),$$

as required.

Deducing $\alpha > 45°$. By (a), $\lambda > 1$, so $\lambda^2 + 4(\lambda - 1) > \lambda^2 > 0$. Write $s = \sqrt{\lambda^2 + 4\lambda - 4}$, so that $s > \lambda$. The inequality in (b) says

$$2t > \lambda + s \qquad \text{or} \qquad 2t < \lambda - s.$$

The second option gives $t < 0$, because $s > \lambda$. But (a) forces $t > 0$: from $(t - \lambda)^2 < \lambda^2 - 1 < \lambda^2$ we get $|t - \lambda| < \lambda$, so $t > 0$. (Physically, the particle must rise to clear the near edge.) Hence the first option holds, and

$$2t > \lambda + s > \lambda + \lambda = 2\lambda > 2,$$

so $\tan\alpha > \lambda > 1$. Since $0 < \alpha < 90°$, this gives $\alpha > 45°$.

Part (ii)

Conversely, the two conditions together are enough. If $y(d) > 0$ then, as $y(0) = 0$ and y is concave, $y > 0$ on the whole interval $0 < x \leq d$, so the particle clears the ground and the near wall. If also $y(2d) < -2d$, it reaches the base at some x strictly between d and $2d$ and stays inside the trench until then. So we need only show that, for every $\lambda > 1$, some $t > 0$ satisfies both inequalities.

From (a), the allowed values are $\lambda - \sqrt{\lambda^2 - 1} < t < \lambda + \sqrt{\lambda^2 - 1}$. From (b), together with $t > 0$, they are $t > \frac{1}{2}(\lambda + s)$. Both hold for some t provided

$$\tfrac{1}{2}(\lambda + s) < \lambda + \sqrt{\lambda^2 - 1} \iff s < \lambda + 2\sqrt{\lambda^2 - 1}.$$

(The lower bound in (a) causes no trouble, since $\frac{1}{2}(\lambda + s) > \lambda > \lambda - \sqrt{\lambda^2 - 1}$.) Both sides are positive, so we may square:

$$\begin{aligned} & \lambda^2 + 4\lambda - 4 < \lambda^2 + 4\lambda\sqrt{\lambda^2 - 1} + 4(\lambda^2 - 1) \\ \iff \quad & 4\lambda < 4\lambda\sqrt{\lambda^2 - 1} + 4\lambda^2 \\ \iff \quad & 1 < \sqrt{\lambda^2 - 1} + \lambda, \end{aligned}$$

dividing by $4\lambda > 0$. The last line is true because $\lambda > 1$. So the interval

$$\tfrac{1}{2}\left(\lambda + \sqrt{\lambda^2 + 4\lambda - 4}\right) < \tan\alpha < \lambda + \sqrt{\lambda^2 - 1}$$

is non-empty. Choosing α with $\tan\alpha$ in it (possible, since tan takes every positive value on $(0^\circ, 90^\circ)$) makes the particle land on the base without touching either side.

Checks

- Take $\lambda = 2$. The interval is $1 + \sqrt{2} < \tan\alpha < 2 + \sqrt{3}$, which is exactly $67.5^\circ < \alpha < 75^\circ$. Trying $\alpha = 70^\circ$ in $y = xt - x^2(1 + t^2)/(4d)$ gives $y(d) \approx 0.61d > 0$ and $y(2d) \approx -3.05d < -2d$, as it should.
- As $\lambda \to 1^+$, both ends of the interval tend to 1, so the window shrinks to $\alpha = 45^\circ$: with $v^2 = dg$ the particle can only just reach the near edge, which needs the maximum-range angle.
- A computer check of thousands of random (λ, α) pairs, stepping the motion forward in time, agreed with the two inequalities in every case.

Common slips

- Treating the conditions as $y(d) > -2d$ (above the base) instead of $y(d) > 0$: the near wall rises all the way to ground level.
- Forgetting to rule out the branch $2\tan\alpha < \lambda - s$ in (b), which needs the fact that $\tan\alpha > 0$.
- Squaring an inequality in (ii) without first noting that both sides are positive.

Beyond STEP

Here a physical condition ("lands in the trench") becomes a region in the space of launch parameters, cut out by inequalities, and the question is whether that region is non-empty. This is the starting point of optimisation and control: describing the set of feasible inputs and finding its boundary. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and its extensions (air resistance, rotating frames) at first-year level.

WORKED SOLUTION

STEP II 2024, Question 10

Mechanics • Friction on Accelerating Wedge

STEP II 2024, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

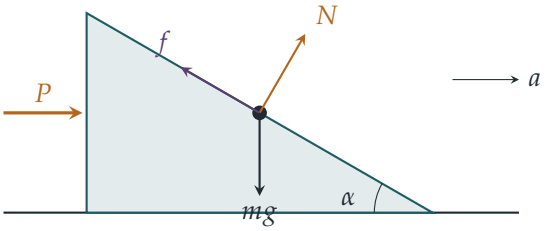
www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything hinges on two observations. First, the plane is smooth, so the only horizontal external force on the whole system (prism plus particle) is P: that gives the common acceleration at once. Second, since the particle does not move relative to the prism, it has exactly that same horizontal acceleration, so resolving for the particle alone along and perpendicular to the slope gives the friction and the normal reaction directly. The inequality in (iii) is then just "friction $\leq \mu\times$ normal reaction", tidied up with the compound-angle formulae.

Solution

We model the particle as a particle, the prism as a rigid body sliding on a smooth horizontal plane, and the contact between particle and oblique face as rough with coefficient μ. Place the vertical face on the left, so that the oblique face slopes down to the right; then P pushes the prism to the right.



On the particle act its weight mg, the normal reaction N perpendicular to the oblique face, and friction, which we take as a signed quantity f acting *up* the slope (if f turns out negative, the friction actually acts down the slope). The magnitude of the friction is $F = |f|$.

Part (i)

Suppose the system is in equilibrium, so everything is at rest. Resolving for the particle perpendicular to and along the slope,

$$N = mg\cos\alpha, \qquad f = mg\sin\alpha.$$

For the particle not to slip we need $|f| \le \mu N$, that is $mg\sin\alpha \le \mu\, mg\cos\alpha$, or $\mu \ge \tan\alpha$ (we can divide by $mg\cos\alpha > 0$ since α is acute). So if $\mu < \tan\alpha$ no equilibrium is possible: the particle must slide.

Part (ii)

The acceleration. Take the prism and particle together. The plane is smooth, so it exerts only a vertical force; the forces between particle and prism are internal. Hence the only horizontal external force is P, and since both bodies share the same acceleration a (horizontal, to the right),

$$P = (M+m)a, \qquad a = \frac{P}{M+m}.$$

Resolving for the particle. The particle's acceleration is a horizontally. Its component up the slope is $-a\cos\alpha$ (the slope rises to the left, and the particle accelerates to the right), and its component along the outward normal is $a\sin\alpha$.

> **Ask yourself** Why resolve along and perpendicular to the slope rather than horizontally and vertically?

Along the slope only f and the weight have components, and perpendicular to it only N and the weight. So each equation contains just one unknown, and we avoid solving simultaneous equations. The price is that we must resolve the acceleration too, which is easy.

Up the slope:

$$f - mg\sin\alpha = -ma\cos\alpha \quad\Longrightarrow\quad f = m(g\sin\alpha - a\cos\alpha).$$

Substituting $a = P/(M+m)$,

$$f = \frac{m}{M+m}[(M+m)g\sin\alpha - P\cos\alpha],$$

and so the magnitude of the friction is

$$F = \frac{m}{M+m}\,|(M+m)g\sin\alpha - P\cos\alpha|,$$

as required. The modulus is genuinely needed: for small P friction acts up the slope, but for large P the slope is pushed into the particle hard enough that friction must act down the slope to stop it riding up.

Perpendicular to the slope (outward normal):

$$N - mg\cos\alpha = ma\sin\alpha \quad \Longrightarrow \quad N = \frac{m}{M+m}[(M+m)g\cos\alpha + P\sin\alpha].$$

This is positive, since $P > 0$ and α is acute, so no modulus is needed and the particle stays in contact.

Part (iii)

The particle stays fixed relative to the prism exactly when friction can supply what is needed: $F \leq \mu N$. Write $G = (M+m)g$ and cancel the positive factor $m/(M+m)$:

$$|G\sin\alpha - P\cos\alpha| \leq \tan\lambda\,(G\cos\alpha + P\sin\alpha).$$

Multiplying by $\cos\lambda > 0$, this is the pair of inequalities

$$-\sin\lambda\,(G\cos\alpha + P\sin\alpha) \;\leq\; \cos\lambda\,(G\sin\alpha - P\cos\alpha) \;\leq\; \sin\lambda\,(G\cos\alpha + P\sin\alpha).$$

? **Ask yourself** How do these turn into tangents of $\alpha \pm \lambda$?

Collect the G terms on one side and the P terms on the other; each bracket is then a compound-angle expansion.

Right-hand inequality (friction up the slope must not exceed its limit):

$$\begin{aligned} G(\sin\alpha\cos\lambda - \cos\alpha\sin\lambda) &\leq P(\cos\alpha\cos\lambda + \sin\alpha\sin\lambda), \\ G\sin(\alpha-\lambda) &\leq P\cos(\alpha-\lambda). \end{aligned}$$

Since $0 < \alpha - \lambda < \frac{1}{4}\pi$, $\cos(\alpha-\lambda) > 0$, so this is $P \geq G\tan(\alpha-\lambda)$.

Left-hand inequality (friction down the slope must not exceed its limit):

$$\begin{aligned} P(\cos\alpha\cos\lambda - \sin\alpha\sin\lambda) &\leq G(\sin\alpha\cos\lambda + \cos\alpha\sin\lambda), \\ P\cos(\alpha+\lambda) &\leq G\sin(\alpha+\lambda). \end{aligned}$$

Here we use the condition $\lambda < \alpha < \frac{1}{4}\pi$: it gives $0 < \alpha + \lambda < \frac{1}{2}\pi$, so $\cos(\alpha+\lambda) > 0$ and we may divide to get $P \leq G\tan(\alpha+\lambda)$.

Both must hold, so

$$(M+m)g\tan(\alpha-\lambda) \leq P \leq (M+m)g\tan(\alpha+\lambda),$$

as required.

Checks

- With $P = 0$ the lower bound says $0 \geq G\tan(\alpha-\lambda)$, which is false because $\lambda < \alpha$: consistent with part (i), since $\mu = \tan\lambda < \tan\alpha$.

- At $P = G\tan\alpha$ we get $f = 0$: the push is exactly right for a smooth particle to sit still, which is the familiar "accelerating wedge" result $a = g\tan\alpha$. This value lies inside the interval, as it should.
- Numerically, with $M = 3$, $m = 1$, $g = 9.8$, $\alpha = 30^\circ$, $\lambda = 10^\circ$, the allowed range is $14.27 \le P \le 32.89$ (newtons). At $P = 20$ the formulae give $F \approx 0.570$ and $N \approx 10.99$, so $F/N \approx 0.052 < \tan 10^\circ \approx 0.176$. Solving the particle's equations directly for thousands of random cases by computer agrees with both formulae and with the interval.

Common slips

- Taking $N = mg\cos\alpha$ in part (ii): the slope is accelerating, so the normal reaction is not the static one.
- Dropping the modulus and finding only the lower bound on P; the upper bound comes from friction acting *down* the slope.
- Dividing by $\cos(\alpha + \lambda)$ without noting that it is positive; this is exactly where $\alpha < \frac{1}{4}\pi$ is used.

Beyond STEP

A cleaner way to see this is to work in the prism's frame, which accelerates at a: there the particle feels an extra "fictitious" force ma backwards, and the problem becomes ordinary statics on a rough slope with an effective gravity tilted by the angle $\arctan(a/g)$. The condition then says that this tilt must lie within λ of α, the angle of friction, which is exactly the interval found. Non-inertial frames and fictitious forces are developed carefully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2024, Question 11

Probability and Statistics • Group Testing Expectation

STEP II 2024, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Pooling only pays off if the group test usually comes back clear, and the expected number of tests per person turns out to be $1 + \frac{1}{k} - (1-p)^k$. Asking when this is at most 1 becomes a comparison between $(1-p)$ and $k^{-1/k}$, and that is exactly why part (i) asks about $x^{1/x}$: the best group size for a high p is the integer that makes $k^{1/k}$ largest.

Solution

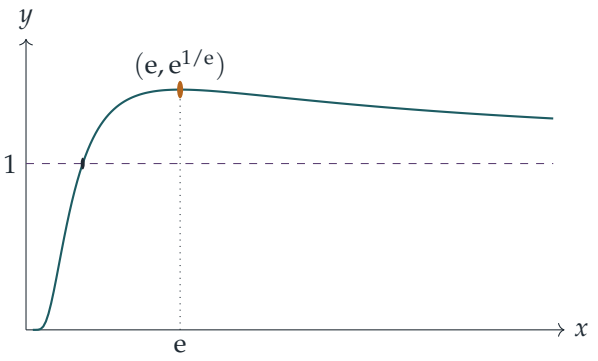
Part (i)

For $x > 0$ write $y = x^{1/x} = \mathrm{e}^{(\ln x)/x}$, so $\ln y = \dfrac{\ln x}{x}$. Differentiating,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1 - \ln x}{x^2}.$$

Since $y > 0$, $\frac{\mathrm{d}y}{\mathrm{d}x}$ has the sign of $1 - \ln x$: positive for $0 < x < \mathrm{e}$, zero at $x = \mathrm{e}$, negative for $x > \mathrm{e}$. So there is a single turning point, a maximum at $(\mathrm{e}, \mathrm{e}^{1/\mathrm{e}}) \approx (2.72,\ 1.44)$.

For the ends of the graph: as $x \to 0^+$, $(\ln x)/x \to -\infty$, so $y \to 0$; as $x \to \infty$, $(\ln x)/x \to 0$, so $y \to 1$ from above (for $x > 1$, $\ln y > 0$). The curve passes through $(1, 1)$.



For positive integers n, the function is increasing on $(0, \mathrm{e}]$ and decreasing on $[\mathrm{e}, \infty)$, so $1^{1/1} < 2^{1/2}$ and $3^{1/3} > 4^{1/4} > 5^{1/5} > \cdots$. The maximum is therefore $2^{1/2}$ or $3^{1/3}$. Raising both to the sixth power gives $2^3 = 8$ and $3^2 = 9$, so $3^{1/3} > 2^{1/2}$ and

$$\max_n n^{1/n} = 3^{1/3}.$$

Part (ii)

Write $q = 1 - p$. Consider one group of k people. By independence, the probability that nobody in the group has the enzyme is q^k, and only then is the pooled sample clear. So the group uses

$$\begin{cases} 1 \text{ test} & \text{with probability } q^k, \\ 1 + k \text{ tests} & \text{with probability } 1 - q^k, \end{cases}$$

and its expected number of tests is $1 + k(1 - q^k)$. There are $r = N/k$ groups, and expectation is additive, so the expected total is

$$E = \frac{N}{k}\left(1 + k(1 - q^k)\right) = N\left(1 + \frac{1}{k} - (1 - p)^k\right).$$

Part (iii)

From part (ii), $E \leq N$ if and only if

$$\frac{1}{k} \leq (1 - p)^k \iff 1 - p \geq k^{-1/k} = \frac{1}{k^{1/k}},$$

taking positive kth roots (both sides are positive, and $t \mapsto t^{1/k}$ is increasing).

Ask yourself Which k allows the largest p?

For a given k the condition is $p \leq 1 - k^{-1/k}$, and this bound is largest when $k^{1/k}$ is largest. By part (i), $k^{1/k} \leq 3^{1/3}$ for every integer $k > 1$, so whatever k is chosen we need $p \leq 1 - 3^{-1/3}$. Conversely, N is a multiple of 3, so $k = 3$ is an allowed group size, and with $k = 3$ the condition is exactly $p \leq 1 - 3^{-1/3}$ (with equality giving $E = N$). Hence

the largest value is

$$p = 1 - 3^{-1/3} \approx 0.307.$$

To show this exceeds $\frac{1}{4}$:

$$1 - 3^{-1/3} > \tfrac{1}{4} \iff 3^{-1/3} < \tfrac{3}{4} \iff \tfrac{1}{3} < \tfrac{27}{64} \iff 64 < 81,$$

cubing in the middle step (cubing preserves order for positive numbers). The last statement is true, so $1 - 3^{-1/3} > \frac{1}{4}$.

Part (iv)

By the binomial expansion,

$$(1-p)^k = 1 - kp + \binom{k}{2}p^2 - \cdots.$$

If pk is small then $\binom{k}{2}p^2 < \frac{1}{2}(pk)^2$ is much smaller again, as are the later terms, so $(1-p)^k \approx 1 - pk$. Substituting into part (ii),

$$E \approx N\left(1 + \frac{1}{k} - (1 - pk)\right) = N\left(\frac{1}{k} + pk\right).$$

(In fact Bernoulli's inequality $(1-p)^k \geq 1 - pk$ shows the approximation never underestimates E, and stopping the expansion one term later shows the error is at most $\frac{1}{2}(pk)^2 N$.)

With $p = 0.01$ and $k = 10$ we have $pk = 0.1$, which is small, and

$$E \approx N\left(\frac{1}{10} + 0.1\right) = 0.2N,$$

so the scheme needs only about 20% of the N tests that testing everyone separately would use.

Checks

- Exactly, with $p = 0.01$ and $k = 10$: $E/N = 1.1 - 0.99^{10} \approx 1.1 - 0.9044 = 0.1956$, about 19.6%, within the error bound $\frac{1}{2}(0.1)^2 = 0.005$ of the approximation.
- At $p = 1 - 3^{-1/3}$ and $k = 3$: $(1-p)^3 = \frac{1}{3}$, so $E = N(1 + \frac{1}{3} - \frac{1}{3}) = N$, exactly on the boundary.
- Sanity of part (ii): as $p \to 0$, $E \to N/k$ (one test per group); as $p \to 1$, $E \to N(1 + \frac{1}{k})$ (every group is retested). The formula was also confirmed by computer, by enumerating all outcomes of small groups.

Common slips

- Answering part (i) with $e^{1/e}$: n must be an integer, so compare $2^{1/2}$ and $3^{1/3}$.

- In part (iii), checking only $k = 3$ without explaining why no other group size could allow a larger p.
- Forgetting the pooled test itself in a positive group, giving k rather than $k + 1$ tests.

Beyond STEP

This is Dorfman's group testing scheme, devised in the Second World War to screen soldiers' blood samples, and it was revived for COVID-19 testing. Minimising $\frac{1}{k} + pk$ gives $k \approx 1/\sqrt{p}$ (so $k = 10$ is optimal here) and about $2\sqrt{p}\,N$ tests; smarter adaptive schemes, studied in information theory, do better still. For the probability behind it, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP II 2024, Question 12

Probability and Statistics • Discrete Order Statistics

STEP II 2024, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

A winner in the first game is a player holding a *unique* smallest number, and there can be at most one such player. So "there is a winner" splits into disjoint cases according to who wins and with which number, and each case is a simple product of independent probabilities. The given formulae for $\sum i^2$ and $\sum i^3$ are the hint that the resulting sums are polynomial sums. The second game works the same way, now with an ordered pair of players (one lowest, one highest) and two numbers to sum over.

Solution

Throughout, the k numbers are independent and each is uniform on $\{1, 2, \ldots, n\}$, so any particular player receives any particular number with probability $\frac{1}{n}$, and receives a number in a given set of s values with probability $\frac{s}{n}$.

Part (i)

Fix a with $1 \le a \le n-1$. Ada must receive a (probability $\frac{1}{n}$), and each of the other $k-1$ players must receive one of the $n-a$ numbers $a+1, \ldots, n$ (probability $\frac{n-a}{n}$ each). By independence,

$$\mathrm{P}(\text{Ada gets } a, \text{ all others larger}) = \frac{1}{n}\left(\frac{n-a}{n}\right)^{k-1} = \frac{(n-a)^{k-1}}{n^k}.$$

(For $a = n$ this would be zero, which is why a stops at $n-1$.)

Let W be the event that there is a winner. A winner holds a number strictly smaller than everyone else's, so there cannot be two winners: each would have to be smaller than the other. Hence W is the disjoint union, over the k players and over $a = 1, \ldots, n-1$, of the events "this player gets a and all the others get larger numbers". By symmetry each

player contributes the same amount as Ada, so

$$\mathrm{P}(W) = k\sum_{a=1}^{n-1}\frac{(n-a)^{k-1}}{n^k} = \frac{k}{n^k}\sum_{j=1}^{n-1} j^{k-1},$$

where we substituted $j = n-a$, which also runs from 1 to $n-1$.

With $k=4$, using the given formula for $\sum i^3$ with $n-1$ in place of n,

$$\mathrm{P}(W) = \frac{4}{n^4}\cdot\frac{1}{4}(n-1)^2n^2 = \frac{(n-1)^2}{n^2},$$

as required.

Part (ii)

Fix d and a with $1 \le d \le n-2$ and $1 \le a \le n-d-1$; these conditions guarantee that $a+d+1 \le n$ and that there is at least one number strictly between a and $a+d+1$. Ada must receive a and Bob must receive $a+d+1$ (probability $\frac{1}{n}$ each), and each of the other $k-2$ players must receive one of the d numbers $a+1,\ldots,a+d$. By independence,

$$\mathrm{P}(\text{Ada gets } a,\ \text{Bob gets } a+d+1,\ \text{others strictly between}) = \frac{1}{n^2}\left(\frac{d}{n}\right)^{k-2} = \frac{d^{k-2}}{n^k}.$$

Ask yourself Why does every way of getting two winners appear exactly once in a sum over players and over a and d?

Two winners means one player has a unique smallest number and a different player has a unique largest number. As in part (i), the low winner is unique and so is the high winner, so the outcome determines an *ordered* pair of distinct players (low winner, high winner) together with the low number a and the gap $d \ge 1$ to the high number: the gap must be at least 1 because the $k-2 \ge 2$ other players' numbers lie strictly between. Different choices of these give disjoint events. There are $k(k-1)$ ordered pairs of players, and each contributes the same as (Ada, Bob), so

$$\mathrm{P}(\text{two winners}) = k(k-1)\sum_{d=1}^{n-2}\sum_{a=1}^{n-d-1}\frac{d^{k-2}}{n^k} = \frac{k(k-1)}{n^k}\sum_{d=1}^{n-2}(n-1-d)\,d^{k-2},$$

since the inner sum has $n-d-1$ equal terms.

Now take $k=4$ and write $m = n-1$, so d runs from 1 to $m-1$. Using both given formulae with $m-1$ in place of n,

$$\begin{aligned}\sum_{d=1}^{m-1}(m-d)\,d^2 &= m\cdot\tfrac{1}{6}(m-1)m(2m-1) - \tfrac{1}{4}(m-1)^2m^2\\ &= \tfrac{1}{12}m^2(m-1)\big[2(2m-1)-3(m-1)\big]\\ &= \tfrac{1}{12}m^2(m-1)(m+1).\end{aligned}$$

Therefore, since $m-1=n-2$ and $m+1=n$,

$$\mathrm{P}(\text{two winners}) = \frac{12}{n^4}\cdot\frac{1}{12}(n-1)^2(n-2)n = \frac{(n-2)(n-1)^2}{n^3},$$

which is the required result.

Comparing one and two winners when $k=4$

Let L be the event that some player has a unique smallest number and H the event that some player has a unique largest number. Part (i) gives $\mathrm{P}(L)=\frac{(n-1)^2}{n^2}$. Replacing every number x by $n+1-x$ turns a uniform number into a uniform number and swaps "smallest" with "largest", so $\mathrm{P}(H)=\mathrm{P}(L)$ too.

Exactly two winners is the event $L\cap H$, so we write $T=\mathrm{P}(L\cap H)=\frac{(n-2)(n-1)^2}{n^3}$. Exactly one winner means L or H happens but not both, so

$$\mathrm{P}(\text{exactly one}) = \mathrm{P}(L)+\mathrm{P}(H)-2\,\mathrm{P}(L\cap H) = \frac{2(n-1)^2}{n^2}-2T.$$

Two winners are more likely than one exactly when

$$\begin{aligned} T > \frac{2(n-1)^2}{n^2}-2T &\iff 3\cdot\frac{(n-2)(n-1)^2}{n^3} > \frac{2(n-1)^2}{n^2} \\ &\iff 3(n-2)>2n \iff n>6, \end{aligned}$$

where we divided by the positive quantity $\frac{(n-1)^2}{n^3}$ (positive because $n\geq 3$). At $n=6$ the two probabilities are equal, so the minimum value is

$$n=7.$$

Checks

- For $n=3$ and $k=4$ there are 81 outcomes. A winner with a 1 needs the other three numbers in $\{2,3\}$: $4\cdot 2^3=32$ outcomes. A winner with a 2 needs three 3s: 4 outcomes. So $\mathrm{P}(W)=\frac{36}{81}=\frac{4}{9}=\frac{(3-1)^2}{3^2}$.
- At $n=6$ both "exactly one" and "exactly two" have probability $\frac{25}{54}\approx 0.463$; at $n=7$ they are $\frac{144}{343}\approx 0.420$ and $\frac{180}{343}\approx 0.525$. These, and the formulae in both parts for $n\leq 8$ and $k=2,3,4$, were confirmed by a computer enumeration of every outcome.
- As $n\to\infty$ ties become rare, so with four players there should almost always be both a unique lowest and a unique highest: indeed $\frac{(n-2)(n-1)^2}{n^3}\to 1$.

Common slips

- Forgetting the factor k in part (i), or the factor $k(k-1)$ in part (ii). In part (ii) the pair of winners is ordered, since one is low and one is high.

- Computing "exactly one winner" as $\mathrm{P}(L) + \mathrm{P}(H) - \mathrm{P}(L \cap H)$, which is "at least one"; the overlap must be subtracted twice.
- Answering $n = 6$: there the two probabilities are equal, not strictly ordered.

Beyond STEP

Splitting an event into disjoint pieces labelled by "who" and "where" is the discrete version of finding the distribution of order statistics, the smallest and largest of a sample. For continuous samples the same argument gives the joint density of the minimum and maximum, and the ties that make this question interesting disappear. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet these ideas.

WORKED SOLUTION

STEP III 2024, Question 1

Pure • Telescoping Series

STEP III 2024, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every sum here is handled the same way: split the term into partial fractions, pick out the pieces that are multiples of $1/r^2$ (these add up to a multiple of S_N), and arrange everything else into differences of the form $f(r) - f(r+1)$, which telescope. Work with finite sums up to N throughout, and only let $N \to \infty$ at the end, when every piece has a known limit.

Solution

Part (i)

Over the common denominator $r^2(r+1)$,

$$\frac{1}{r+1} - \frac{1}{r} + \frac{1}{r^2} = \frac{r^2 - r(r+1) + (r+1)}{r^2(r+1)} = \frac{r^2 - r^2 - r + r + 1}{r^2(r+1)} = \frac{1}{r^2(r+1)},$$

as required.

Now sum from $r = 1$ to N. The $1/r^2$ terms give S_N, and the other two terms telescope:

$$\sum_{r=1}^{N} \left(\frac{1}{r+1} - \frac{1}{r} \right) = \left(\tfrac{1}{2} - 1 \right) + \left(\tfrac{1}{3} - \tfrac{1}{2} \right) + \cdots + \left(\tfrac{1}{N+1} - \tfrac{1}{N} \right) = \frac{1}{N+1} - 1.$$

Hence

$$\sum_{r=1}^{N} \frac{1}{r^2(r+1)} = S_N - 1 + \frac{1}{N+1}.$$

As $N \to \infty$, $S_N \to \frac{1}{6}\pi^2$ and $\frac{1}{N+1} \to 0$, so the right-hand side tends to $\frac{1}{6}\pi^2 - 1$. The infinite sum is by definition the limit of the partial sums, so

$$\sum_{r=1}^{\infty} \frac{1}{r^2(r+1)} = \tfrac{1}{6}\pi^2 - 1.$$

Part (ii)

Write

$$\frac{1}{r^2(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r^2} + \frac{C}{r+1} + \frac{D}{r+2}.$$

Multiplying through by $r^2(r+1)(r+2)$ and using the cover-up rule: putting $r = 0$ gives $B = \frac{1}{2}$; $r = -1$ gives $C = \frac{1}{(-1)^2 \cdot 1} = 1$; $r = -2$ gives $D = \frac{1}{4 \cdot (-1)} = -\frac{1}{4}$. For large r the left-hand side is of order $1/r^4$, so the $1/r$ terms on the right must cancel: $A + C + D = 0$, giving $A = -\frac{3}{4}$. So

$$\frac{1}{r^2(r+1)(r+2)} = \frac{1}{2r^2} - \frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)}.$$

Ask yourself The three terms $-\frac{3}{4r}$, $\frac{1}{r+1}$, $-\frac{1}{4(r+2)}$ don't look like a single telescoping difference. How can we make them telescope?

Split the middle coefficient $1 = \frac{3}{4} + \frac{1}{4}$, so that each piece pairs with a neighbour:

$$-\frac{3}{4r} + \frac{1}{r+1} - \frac{1}{4(r+2)} = -\frac{3}{4}\left(\frac{1}{r} - \frac{1}{r+1}\right) + \frac{1}{4}\left(\frac{1}{r+1} - \frac{1}{r+2}\right).$$

Each bracket now telescopes:

$$\sum_{r=1}^{N}\left(\frac{1}{r} - \frac{1}{r+1}\right) = 1 - \frac{1}{N+1}, \qquad \sum_{r=1}^{N}\left(\frac{1}{r+1} - \frac{1}{r+2}\right) = \frac{1}{2} - \frac{1}{N+2}.$$

Therefore

$$\begin{aligned}\sum_{r=1}^{N}\frac{1}{r^2(r+1)(r+2)} &= \tfrac{1}{2}S_N - \tfrac{3}{4}\left(1 - \frac{1}{N+1}\right) + \tfrac{1}{4}\left(\frac{1}{2} - \frac{1}{N+2}\right)\\ &= \tfrac{1}{2}S_N - \frac{5}{8} + \frac{3}{4(N+1)} - \frac{1}{4(N+2)}.\end{aligned}$$

Letting $N \to \infty$, the last two terms tend to 0 and $S_N \to \frac{1}{6}\pi^2$, so

$$\sum_{r=1}^{\infty}\frac{1}{r^2(r+1)(r+2)} = \frac{\pi^2}{12} - \frac{5}{8}.$$

Part (iii)

The target has $\frac{2}{r^2(r+1)}$ on the right, so compare the two terms. Using part (i),

$$\frac{2}{r^2(r+1)} = \frac{2}{r+1} - \frac{2}{r} + \frac{2}{r^2}.$$

On the other hand,

$$\frac{1}{r^2(r+1)^2} = \left(\frac{1}{r} - \frac{1}{r+1}\right)^2 = \frac{1}{r^2} + \frac{1}{(r+1)^2} - \frac{2}{r(r+1)} = \frac{1}{r^2} + \frac{1}{(r+1)^2} - \frac{2}{r} + \frac{2}{r+1},$$

where we used $\frac{1}{r} - \frac{1}{r+1} = \frac{1}{r(r+1)}$ in both steps. Subtracting,

$$\frac{1}{r^2(r+1)^2} - \frac{2}{r^2(r+1)} = \frac{1}{(r+1)^2} - \frac{1}{r^2},$$

that is,

$$\frac{1}{r^2(r+1)^2} = \frac{2}{r^2(r+1)} - \left(\frac{1}{r^2} - \frac{1}{(r+1)^2}\right).$$

Ask yourself Why not just add infinite series term by term straight away?

Rearranging infinite sums is only safe once we know each piece converges, so we sum to N first. The bracket telescopes:

$$\sum_{r=1}^{N} \frac{1}{r^2(r+1)^2} = \sum_{r=1}^{N} \frac{2}{r^2(r+1)} - \left(1 - \frac{1}{(N+1)^2}\right).$$

By part (i), the first sum on the right converges (to $2(\frac{1}{6}\pi^2 - 1)$) as $N \to \infty$, and $\frac{1}{(N+1)^2} \to 0$. So the left-hand side converges too, and

$$\sum_{r=1}^{\infty} \frac{1}{r^2(r+1)^2} = \sum_{r=1}^{\infty} \frac{2}{r^2(r+1)} - 1,$$

as required. Both sides equal $\frac{1}{3}\pi^2 - 3$.

Checks

- Take $N = 1$ in part (ii): the sum is the single term $\frac{1}{6}$, and the formula gives $\frac{1}{2} - \frac{5}{8} + \frac{3}{8} - \frac{1}{12} = \frac{12-15+9-2}{24} = \frac{1}{6}$.
- Numerically, $\frac{1}{6}\pi^2 - 1 \approx 0.644934$, $\frac{1}{12}\pi^2 - \frac{5}{8} \approx 0.197467$ and $\frac{1}{3}\pi^2 - 3 \approx 0.289868$. Adding the first million terms of each series by computer reproduces all three values to at least nine decimal places.
- The finite identities in all three parts were confirmed exactly, with rational arithmetic, for every N up to 59.

Common slips

- Getting the end terms of a telescoping sum wrong, especially for the $\frac{1}{r+1} - \frac{1}{r+2}$ sum, which starts at $\frac{1}{2}$, not 1.
- Splitting into separate infinite series such as $\sum \frac{1}{r}$, which diverge, instead of keeping the harmonic-type pieces together in finite sums.

Beyond STEP

The value $\sum 1/r^2 = \pi^2/6$ is Euler's solution of the Basel problem, usually proved at university with Fourier series or with the product formula for $\sin x$. The care taken here over finite sums is exactly what analysis courses make precise: a series is the limit of its partial sums, and sums of convergent series may be combined, while divergent ones may not. Abbott's *Understanding Analysis* treats series this way and is a good next read.

WORKED SOLUTION

STEP III 2024, Question 2

Pure • Inequalities and Modulus

STEP III 2024, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every expression under a square root here is a quadratic that is always positive, and both sides of each inequality are non-negative, so we may square without changing the solution set. Squaring turns each problem into a statement about the sign of a quadratic (or, when the x^2 terms cancel, a linear function). Geometrically, $y = \sqrt{4x^2 - 8x + 64}$ is the top half of a hyperbola, and the lines in the question are chords and asymptotes of it.

Solution

Part (i)

Completing the square, $4x^2 - 8x + 64 = 4(x-1)^2 + 60 > 0$, so the square root is defined for every x. Both sides of each inequality are non-negative, and for $A, B \geq 0$ we have $A \leq B \iff A^2 \leq B^2$.

(a) Squaring,

$$\begin{aligned} 4x^2 - 8x + 64 \leq x^2 + 16x + 64 &\iff 3x^2 - 24x \leq 0 \\ &\iff 3x(x-8) \leq 0 \iff 0 \leq x \leq 8. \end{aligned}$$

(b) Squaring,

$$\begin{aligned} 4x^2 - 8x + 64 \leq 9x^2 - 48x + 64 &\iff 0 \leq 5x^2 - 40x \\ &\iff 5x(x-8) \geq 0 \iff x \leq 0 \text{ or } x \geq 8. \end{aligned}$$

In both parts, equality holds at $x = 0$ (all sides equal 8) and at $x = 8$ (all sides equal 16).

Part (ii)

(a) Using the difference of two squares,

$$\left(\sqrt{4x^2 - 8x + 64} + 2(x-1)\right)\mathrm{f}(x) = (4x^2 - 8x + 64) - 4(x-1)^2 = 60.$$

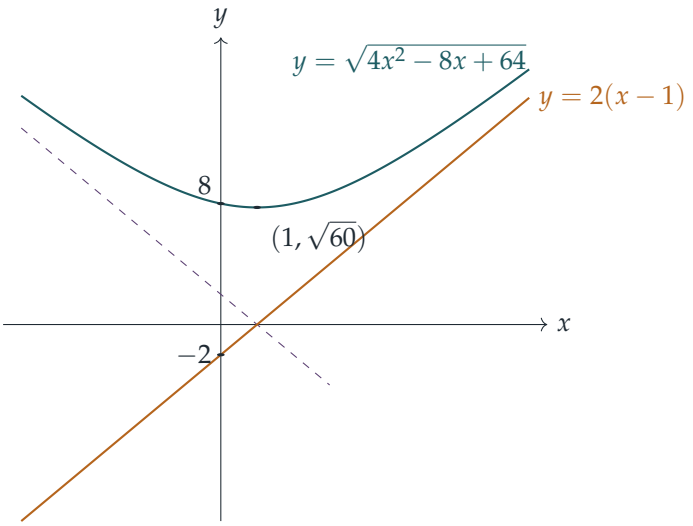
For $x > 1$ the bracket is positive, so we may divide:

$$f(x) = \frac{60}{\sqrt{4x^2 - 8x + 64} + 2(x-1)}.$$

As $x \to \infty$ the denominator is at least $2(x-1)$, which tends to infinity, so $f(x) \to 0$, as required.

(b) Write $y = \sqrt{4(x-1)^2 + 60}$. Then $y > 0$ and $y^2 - 4(x-1)^2 = 60$, so the curve is the upper branch of a hyperbola centred at $(1, 0)$. It is symmetric about $x = 1$, with minimum value $\sqrt{60} = 2\sqrt{15} \approx 7.75$ at $x = 1$, and it crosses the y-axis at $(0, 8)$.

Since $y > \sqrt{4(x-1)^2} = 2|x-1|$, the curve lies strictly above both lines $y = \pm 2(x-1)$. Part (a) says it approaches $y = 2(x-1)$ as $x \to \infty$. By the symmetry about $x = 1$, it approaches $y = -2(x-1)$ as $x \to -\infty$. The line $y = 2(x-1)$ has gradient 2 and passes through $(1, 0)$ and $(0, -2)$.



The dashed line is the other asymptote, $y = -2(x-1)$.

Part (iii)

Again $4x^2 - 5x + 4 = 4\left(x - \frac{5}{8}\right)^2 + \frac{39}{16} > 0$, so the square root is always defined. Squaring both (non-negative) sides, the inequality is equivalent to

$$(m^2 - 4)x^2 + (2mc + 5)x + (c^2 - 4) \geq 0. \qquad (*)$$

> **Ask yourself** What kind of expression on the left of $(*)$ could be non-negative exactly when $x \geq 3$?

If $m^2 - 4 > 0$, the left-hand side tends to $+\infty$ as $x \to -\infty$, so very negative x would satisfy $(*)$. If $m^2 - 4 < 0$, it tends to $-\infty$ as $x \to +\infty$, so large x would fail. Neither is

allowed, so $m^2 = 4$, and $(*)$ becomes linear. A linear inequality $ax + b \geq 0$ has solution set $\{x : x \geq 3\}$ exactly when $a > 0$ and $3a + b = 0$.

Take $m = 2$. Then $a = 4c + 5$ and $b = c^2 - 4$, so we need

$$3(4c + 5) + c^2 - 4 = c^2 + 12c + 11 = (c + 1)(c + 11) = 0, \qquad 4c + 5 > 0.$$

The root $c = -11$ gives $a = -39 < 0$, which is no good. The root $c = -1$ gives $a = 1 > 0$. So

$$m = 2, \qquad c = -1$$

works. To confirm, $(2x - 1)^2 - (4x^2 - 5x + 4) = x - 3$, which is non-negative exactly when $x \geq 3$. (Taking $m = -2$, $c = 1$ gives the same $|mx + c|$ and is equally valid.)

Part (iv)

Write $g(x) = x^2 + px + q$ and $L(x) = mx + c$. Since $|g| \leq L \iff -L \leq g \leq L$, the inequality holds exactly when both

$$g(x) - L(x) \leq 0 \qquad \text{and} \qquad g(x) + L(x) \geq 0.$$

Both $g - L$ and $g + L$ are quadratics with leading coefficient 1.

- The set where $g - L \leq 0$ is a closed interval $[a, b]$ (possibly empty or a single point).
- The set where $g + L \geq 0$ is everything except an open interval (r, s). If $g + L$ has no real roots, nothing is removed.

So the solution set is $[a, b]$ with (r, s) removed. For this to be $[-5, 1] \cup [5, 7]$ we need $a = -5$ and $b = 7$ (the outer ends), and the removed gap must be exactly $(1, 5)$, so $r = 1$ and $s = 5$. Hence

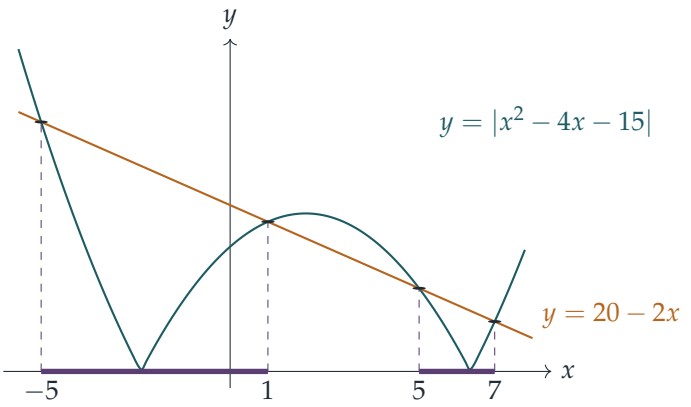
$$g(x) - L(x) = (x + 5)(x - 7) = x^2 - 2x - 35,$$
$$g(x) + L(x) = (x - 1)(x - 5) = x^2 - 6x + 5.$$

Adding and halving gives $g(x) = x^2 - 4x - 15$. Subtracting and halving gives $L(x) = -2x + 20$. Therefore

$$p = -4, \qquad q = -15, \qquad m = -2, \qquad c = 20.$$

Every step above is reversible, so with these values the solution set really is $[-5, 1] \cup [5, 7]$: we need $(x + 5)(x - 7) \leq 0$ and $(x - 1)(x - 5) \geq 0$. These values are also the only ones that work.

In the picture, g has roots $2 \pm \sqrt{19} \approx -2.36,\ 6.36$. The line $y = 20 - 2x$ meets the outer arms of $|g|$ at $x = -5$ and $x = 7$, and cuts through the reflected hump at $x = 1$ and $x = 5$.



Checks

- In (ii)(a), at $x = 100$ we get $f(100) = \sqrt{39264} - 198 \approx 0.151$, matching $60/(\sqrt{39264} + 198) \approx 0.151$.
- In (iv), at $x = 0$: $|-15| = 15 \leq 20$, so 0 is a solution. At $x = 3$: $|9 - 12 - 15| = 18 > 14$, so 3 is not. At $x = 8$: $|64 - 32 - 15| = 17 > 4$, so 8 is not. These match $[-5, 1] \cup [5, 7]$. A computer check over a fine grid of x confirmed all four solution sets.

Common slips

- Squaring without saying why it is allowed. It is valid here only because both sides are non-negative.
- In (iii), keeping a genuine quadratic. A quadratic inequality can never have a one-sided half-line as its solution set, so the x^2 terms must cancel.
- In (iv), pairing the endpoints wrongly: the outer ends $-5, 7$ belong to $g - L$, and the inner ends $1, 5$ (the gap) to $g + L$.

Beyond STEP

The curve $y^2 - 4(x-1)^2 = 60$ is a hyperbola, and part (ii) shows how an asymptote is found: rationalise the difference between the curve and the candidate line. The same trick, multiplying by a "conjugate", is a standard tool for limits in first-year analysis. Writing $|g| \leq L$ as two one-sided conditions, as in (iv), is how inequalities with absolute values are routinely handled there. Abbott's *Understanding Analysis* is a good next step.

WORKED SOLUTION

STEP III 2024, Question 3

Pure • Monotonic Functions and Derivatives

STEP III 2024, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything hangs on one derivative. When you differentiate g and put it over a common denominator, the numerator collapses to a *linear* expression in x, so its sign is easy to read off. The link to part (iii) is that g is exactly the derivative of $\ln f$. So the sign of g, which the sketches in (ii) make visible, decides whether f rises or falls.

Solution

Part (i)

Since $x > 0$, write $\ln\left(1+\frac{1}{x}\right) = \ln(x+1) - \ln x$. Then

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(1+\frac{1}{x}\right) = \frac{1}{x+1} - \frac{1}{x} = -\frac{1}{x(x+1)} = -\frac{x(x+1)}{x^2(x+1)^2},$$

and, by the quotient rule with denominator x^2+x,

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{x+c}{x^2+x} = \frac{(x^2+x)-(x+c)(2x+1)}{x^2(x+1)^2} = \frac{-x^2-2cx-c}{x^2(x+1)^2}.$$

Subtracting, the x^2 terms cancel:

$$\mathrm{g}'(x) = \frac{-x^2-x+x^2+2cx+c}{x^2(x+1)^2} = \frac{(2c-1)x+c}{x^2(x+1)^2}.$$

The denominator is positive, so the sign of $\mathrm{g}'(x)$ is the sign of $N(x) = (2c-1)x+c$.

(a) If $c \geq \frac{1}{2}$ then $(2c-1)x \geq 0$ and $c \geq \frac{1}{2} > 0$, so $N(x) > 0$ for every $x > 0$. Hence the gradient is positive for all $x > 0$.

(b) If $0 \leq c < \frac{1}{2}$ then $1-2c > 0$, and

$$N(x) < 0 \iff (1-2c)x > c \iff x > \frac{c}{1-2c}.$$

So the gradient is negative exactly when $x > \dfrac{c}{1-2c}$. (When $c = 0$ this is every $x > 0$.) Write $x_0 = \dfrac{c}{1-2c}$. For $c > 0$, g is increasing on $(0, x_0]$ and decreasing on $[x_0, \infty)$.

Part (ii)

We also need the behaviour for large x. Both terms of g tend to 0, so $\mathrm{g}(x) \to 0$, but the sketch should show which side it approaches from. Using $\ln(1+t) = t - \frac{1}{2}t^2 + \cdots$ with $t = \frac{1}{x}$, and $\dfrac{x+c}{x(x+1)} = \left(\dfrac{1}{x} + \dfrac{c}{x^2}\right)\left(1 - \dfrac{1}{x} + \cdots\right) = \dfrac{1}{x} + \dfrac{c-1}{x^2} + \cdots$, we get

$$\mathrm{g}(x) = \frac{\frac{1}{2} - c}{x^2} + \cdots \quad \text{for large } x.$$

Ask yourself If g is increasing and tends to 0, which side of the axis must it be on?

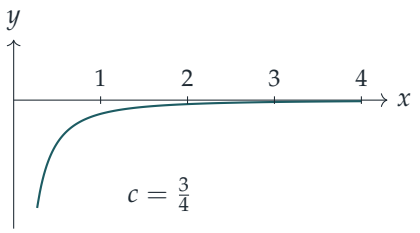
It must be below. If g is strictly increasing then, for any x, $\mathrm{g}(x) < \mathrm{g}(y)$ for all $y > x$, and letting $y \to \infty$ gives $\mathrm{g}(x) < 0$ (strict, because $\mathrm{g}(x) < \mathrm{g}(x+1) \leq 0$). In the same way, a strictly decreasing function that tends to 0 is positive.

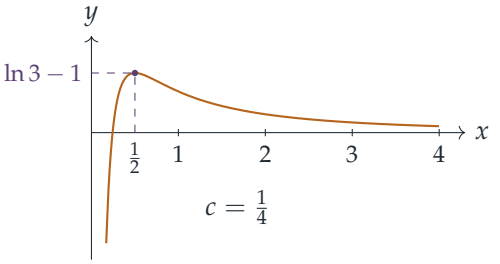
(a) $c = \frac{3}{4}$**.** By (i)(a), g is increasing throughout. It rises from $-\infty$ (as $x \to 0$) towards 0 and stays below the axis. It approaches like $-\frac{1}{4x^2}$, so it never crosses and has no turning point. For example, $\mathrm{g}(1) = \ln 2 - \frac{7}{8} \approx -0.18$.

(b) $c = \frac{1}{4}$**.** Here $x_0 = \dfrac{1/4}{1/2} = \frac{1}{2}$. The curve rises from $-\infty$ to a maximum at $x = \frac{1}{2}$, where

$$\mathrm{g}\left(\tfrac{1}{2}\right) = \ln 3 - \frac{3/4}{3/4} = \ln 3 - 1 \approx 0.099 > 0.$$

It then decreases towards 0 from above, like $\frac{1}{4x^2}$. On $[\frac{1}{2}, \infty)$ it is decreasing to 0, so it is positive there. On $(0, \frac{1}{2})$ it is increasing from $-\infty$ to a positive value, so it crosses the axis exactly once, at a point a little below $\frac{1}{4}$ (numerically $x \approx 0.243$).





(The vertical scales differ: the right-hand graph is stretched to make the small positive hump visible. In both graphs the y-axis is an asymptote.)

Part (iii)

Since $f(x) > 0$ for $x > 0$, we can take logarithms:

$$\ln f(x) = (x+c)\ln\left(1+\frac{1}{x}\right).$$

Differentiating with the product rule, and using the derivative from (i),

$$\frac{f'(x)}{f(x)} = \ln\left(1+\frac{1}{x}\right) + (x+c)\cdot\left(-\frac{1}{x(x+1)}\right) = g(x).$$

So $f'(x) = f(x)\,g(x)$, and because $f(x) > 0$, f' has the same sign as g. Note also that $g(x) \to 0$ as $x \to \infty$ for every c, since both of its terms tend to 0.

(a) $c \geq \frac{1}{2}$**.** By (i)(a), g is strictly increasing on $x > 0$ and tends to 0. By the argument after the box in (ii), $g(x) < 0$ for all $x > 0$. Hence $f'(x) < 0$, and f is decreasing.

(b) $0 < c < \frac{1}{2}$**.** By (i)(b), g is strictly decreasing on $[x_0, \infty)$ and tends to 0, so $g > 0$ there; in particular $g(x_0) > 0$. On $(0, x_0]$, g is continuous and strictly increasing, with $g(x) \to -\infty$ as $x \to 0$ (given, as $c > 0$). It therefore takes the value 0 at exactly one point $x_1 \in (0, x_0)$, with $g < 0$ on $(0, x_1)$ and $g > 0$ on (x_1, ∞). So $f'(x_1) = 0$, and f' changes sign from negative to positive there. Hence f has a turning point (a minimum) at x_1.

> **Ask yourself** Why isn't $f'(x_1) = 0$ on its own enough?

A stationary point need not be a turning point (think of $y = x^3$ at the origin). What makes x_1 a turning point is the change of sign of f', and that is exactly what the shape of g supplies.

(c) $c = 0$**.** Now $g'(x) = \dfrac{-x}{x^2(x+1)^2} < 0$ for all $x > 0$, so g is strictly decreasing and tends to 0. Hence $g(x) > 0$ for all $x > 0$, so $f'(x) > 0$, and $f(x) = \left(1+\frac{1}{x}\right)^x$ is increasing.

Checks

- Differentiate g numerically at a point, say $x = 1$, $c = \frac{1}{4}$: the formula gives $g'(1) = \frac{-1/2+1/4}{4} = -\frac{1}{16}$, which matches a difference quotient.
- $c = \frac{1}{4}$: $g(0.2) \approx -0.083$ and $g(0.3) \approx 0.056$, consistent with the single crossing near 0.243. For $c = \frac{3}{4}$, $g(3) \approx -0.025$, still just below the axis.
- In (iii)(b) with $c = \frac{1}{4}$, the minimum of f is at the same $x \approx 0.243$, and a computer check of f on a fine grid finds exactly one turning point for several values of c in $(0, \frac{1}{2})$.
- For $c = 0$, $f(x) = (1+\frac{1}{x})^x$ increases to e; for $c = 1$, $(1+\frac{1}{x})^{x+1}$ decreases to e. That's the familiar pair of sequences that squeeze e.

Common slips

- Concluding $\mathrm{g} < 0$ from "g increasing" alone. You also need the limit $\mathrm{g} \to 0$ as $x \to \infty$.
- In (ii)(b), drawing the curve as always negative. The maximum $\ln 3 - 1$ is small but positive, so the curve must cross the axis and come back down to it from above.
- Differentiating $(1 + \frac{1}{x})^{x+c}$ as if the exponent were constant. Take logarithms first.

Beyond STEP

The fact that $(1 + \frac{1}{x})^x$ increases and $(1 + \frac{1}{x})^{x+1}$ decreases, both towards e, is a classic first-year way of proving that the limit defining e exists. It is an instance of the monotone convergence theorem. This question finds the precise threshold $c = \frac{1}{2}$ where the behaviour switches, and the expansion $\mathrm{g}(x) \approx (\frac{1}{2} - c)/x^2$ explains why. For more on monotone sequences and limits, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2024, Question 4

Pure • Tangents to a Parabola

STEP III 2024, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything depends on the formula for tan of the angle between two lines, which is just the $\tan(A-B)$ formula in disguise. Once the tangents at $x=p$ and $x=q$ are known to meet at a point (X,Y) with $p+q$ and pq given in terms of X and Y, any condition on the angle, which involves $(p-q)^2$ and pq, can be rewritten in X and Y alone. That rewritten condition is the locus.

Solution

Part (i)

Let the lines make angles α and β with the positive x-axis, so $m_1=\tan\alpha$ and $m_2=\tan\beta$ (both lines have gradients, so neither is vertical). The angle between the lines is $\alpha-\beta$ up to sign and up to adding multiples of 180°. If the acute angle between them is 45°, then $\alpha-\beta$ is $\pm 45^\circ$ plus a multiple of 180°, and since tan has period 180°,

$$\tan(\alpha-\beta)=\tan(\pm 45^\circ)=\pm 1.$$

In particular $\cos(\alpha-\beta)\neq 0$, so the compound-angle formula applies and $1+m_1m_2\neq 0$:

$$\tan(\alpha-\beta)=\frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}=\frac{m_1-m_2}{1+m_1m_2}.$$

Hence $\dfrac{m_1-m_2}{1+m_1m_2}=\pm 1$, as required. More generally, the acute angle θ between the lines satisfies $\tan\theta=\left|\dfrac{m_1-m_2}{1+m_1m_2}\right|$, which we use in part (iii).

Part (ii)

On C we have $y = \dfrac{x^2}{4a}$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x}{2a}$. The tangent at $\left(p, \frac{p^2}{4a}\right)$ is

$$y - \frac{p^2}{4a} = \frac{p}{2a}(x - p) \quad \Longleftrightarrow \quad y = \frac{p}{2a}x - \frac{p^2}{4a},$$

and similarly the tangent at $x = q$ is $y = \dfrac{q}{2a}x - \dfrac{q^2}{4a}$. Subtracting,

$$\frac{p-q}{2a}\,x = \frac{p^2 - q^2}{4a} = \frac{(p-q)(p+q)}{4a}.$$

Since $p \neq q$ we may divide by $p - q$, giving $x = \frac{1}{2}(p+q)$, as required. Substituting into the first tangent,

$$y = \frac{p(p+q)}{4a} - \frac{p^2}{4a} = \frac{pq}{4a}.$$

So the tangents meet at

$$(X, Y) = \left(\tfrac{1}{2}(p+q),\ \frac{pq}{4a}\right), \qquad \text{so} \qquad p + q = 2X, \quad pq = 4aY. \tag{$*$}$$

Now suppose the tangents at $x = p$ and $x = q$ are at 45° to each other. They are certainly different lines, so $p \neq q$ and $(*)$ holds. Their gradients are $m_1 = \frac{p}{2a}$ and $m_2 = \frac{q}{2a}$, so

$$\frac{m_1 - m_2}{1 + m_1 m_2} = \frac{(p-q)/(2a)}{1 + pq/(4a^2)} = \frac{2a(p-q)}{4a^2 + pq}.$$

By part (i) this is ± 1; squaring removes the sign:

$$4a^2(p-q)^2 = (4a^2 + pq)^2.$$

> **Ask yourself** How do we get rid of p and q when the condition involves $p - q$, not $p + q$?

Use $(p-q)^2 = (p+q)^2 - 4pq$, which is symmetric, so $(*)$ applies:

$$(p-q)^2 = 4X^2 - 16aY, \qquad 4a^2 + pq = 4a^2 + 4aY = 4a(a + Y).$$

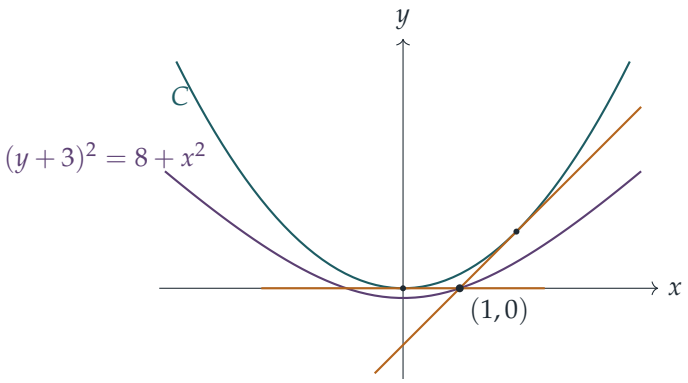
The condition becomes

$$4a^2\,(4X^2 - 16aY) = 16a^2(a + Y)^2.$$

Dividing by $16a^2$ (allowed as $a \neq 0$),

$$\begin{aligned} & X^2 - 4aY = Y^2 + 2aY + a^2, \\ \text{so} \qquad & X^2 + 8a^2 = Y^2 + 6aY + 9a^2 = (Y + 3a)^2. \end{aligned}$$

So the meeting point lies on $(y + 3a)^2 = 8a^2 + x^2$, as required.



With $a = 1$: the tangents at $x = 2$ (gradient 1) and $x = 0$ (gradient 0) are at 45°, and they meet at $(1, 0)$, which is on the upper branch of the hyperbola.

Part (iii)

Let (X, Y) be a point on $(y + 7a)^2 = 48a^2 + 3x^2$ where two tangents to C, at $x = p$ and $x = q$ with $p \neq q$, meet, so $(*)$ holds. By the remark at the end of part (i), the acute angle θ between them satisfies

$$\tan^2\theta = \left(\frac{2a(p-q)}{4a^2+pq}\right)^2 = \frac{4a^2(4X^2-16aY)}{16a^2(a+Y)^2} = \frac{X^2-4aY}{(Y+a)^2},$$

using the same substitutions as in part (ii). (If $Y = -a$ the tangents would be perpendicular; we show below that this cannot happen on this curve.)

? **Ask yourself** How does the equation of the curve help with $X^2 - 4aY$?

Expand the curve's equation: $Y^2 + 14aY + 49a^2 = 48a^2 + 3X^2$, that is,

$$3X^2 = Y^2 + 14aY + a^2.$$

Then

$$3(X^2 - 4aY) = Y^2 + 14aY + a^2 - 12aY = Y^2 + 2aY + a^2 = (Y + a)^2.$$

Two consequences follow. First, $Y \neq -a$: otherwise $X^2 - 4aY = 0$, i.e. $X^2 = -4a^2$, impossible as $a \neq 0$. So $X^2 - 4aY = \frac{1}{3}(Y + a)^2 > 0$, which also confirms that the point lies outside C: the quadratic $t^2 - 2Xt + 4aY = 0$, whose roots are p and q, has two distinct real roots, so there really are two tangents through every point of the curve. Second,

$$\tan^2\theta = \frac{\frac{1}{3}(Y+a)^2}{(Y+a)^2} = \frac{1}{3}.$$

So $\tan\theta = \frac{1}{\sqrt{3}}$ wherever the point is on the curve: the acute angle between the tangents is constant, and it equals 30°.

Checks

- With $a = 1$, the tangents at $x = 2$ and $x = 0$ are $y = x - 1$ and $y = 0$, at $45°$. They meet at $(1, 0)$, and $(0 + 3)^2 = 9 = 8 + 1^2$.
- For part (iii) with $a = 1$, take $X = 0$: then $(Y + 7)^2 = 48$, so $Y = 4\sqrt{3} - 7 \approx -0.072$. The tangents through $(0, Y)$ touch at $x = \pm\sqrt{-4Y} \approx \pm 0.536$, with gradients ± 0.268. Since $\tan 15° \approx 0.268$, each makes $15°$ with the horizontal and they are $30°$ apart.
- A computer check over a thousand random values of a and points on both branches of each hyperbola confirmed the $45°$ and $30°$ results.

Common slips

- Trying to solve for p and q separately. Only the symmetric combinations $p + q$ and pq are needed, with $(p - q)^2 = (p + q)^2 - 4pq$.
- Forgetting the $\pm$ in part (i): squaring the condition keeps both orientations of the $45°$ angle, and that is what gives a single equation.
- In part (iii), not checking that $Y + a \neq 0$ before cancelling.

Beyond STEP

The tangents from a point (X, Y) to the parabola are governed by the quadratic $t^2 - 2Xt + 4aY = 0$, and the locus of points from which the tangents meet at a fixed angle is always a hyperbola (for $90°$ it degenerates to the directrix $y = -a$). These ideas belong to projective geometry, where the pole–polar relation links a point to its chord of contact. For more questions in this spirit, with commentary on how to attack them, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2024, Question 5

Pure • Matrix Differential Equations

STEP III 2024, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.7, *Quantities That Never Change*: https://beyondstep.keithmcnulty.org/ch07/lax-pairs.html

The key idea

Everything hangs on two facts about the trace: it is linear, and $\operatorname{tr}(\mathbf{XY}) = \operatorname{tr}(\mathbf{YX})$. Part (ii) is a formula for the rate of change of the determinant (Jacobi's formula), and it turns "is $\det \mathbf{M}$ constant?" into "is a certain trace zero?". In (iii) a commutator $\mathbf{MN} - \mathbf{NM}$ always has trace zero, which gives all three invariants. In (iv) the same formula shows that $\operatorname{tr} \mathbf{N}$ measures how fast $\ln|\det \mathbf{M}|$ changes, so the question becomes whether a matrix can have constant trace but varying determinant.

Solution

Part (i)

Write $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. The diagonal entries of $\mathbf{MN}$ are $ap + br$ and $cq + ds$, while those of $\mathbf{NM}$ are $pa + qc$ and $rb + sd$. So

$$\operatorname{tr}(\mathbf{MN}) = ap + br + cq + ds = \operatorname{tr}(\mathbf{NM}).$$

Also $\mathbf{M} + \mathbf{N}$ has diagonal entries $a + p$ and $d + s$, so

$$\operatorname{tr}(\mathbf{M} + \mathbf{N}) = \operatorname{tr}(\mathbf{M}) + \operatorname{tr}(\mathbf{N}).$$

In the same way $\operatorname{tr}(k\mathbf{M}) = k \operatorname{tr}(\mathbf{M})$ for a scalar k, and the trace of $\mathrm{d}\mathbf{M}/\mathrm{d}t$ is the derivative of $\operatorname{tr} \mathbf{M}$. We use these freely below.

Part (ii)

With entries a, b, c, d depending on t (dashes denote $\mathrm{d}/\mathrm{d}t$), the product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = \frac{\mathrm{d}}{\mathrm{d}t}(ad - bc) = a'd + ad' - b'c - bc'.$$

Since $\mathbf{M}$ is non-singular, $\mathbf{M}^{-1} = \dfrac{1}{\det \mathbf{M}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, and

$$\mathbf{M}^{-1}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \frac{1}{\det \mathbf{M}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \frac{1}{\det \mathbf{M}} \begin{pmatrix} da' - bc' & * \\ * & -cb' + ad' \end{pmatrix}.$$

Its trace is $\dfrac{a'd + ad' - b'c - bc'}{\det \mathbf{M}}$, which is $\dfrac{1}{\det \mathbf{M}} \dfrac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M})$, as required.

Part (iii)

Determinant. By (ii) and linearity of the trace,

$$\frac{1}{\det \mathbf{M}} \frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = \operatorname{tr}\big(\mathbf{M}^{-1}(\mathbf{MN} - \mathbf{NM})\big) = \operatorname{tr}(\mathbf{N}) - \operatorname{tr}(\mathbf{M}^{-1}\mathbf{NM}).$$

Using (i) with the two matrices $\mathbf{M}^{-1}\mathbf{N}$ and $\mathbf{M}$, we get $\operatorname{tr}(\mathbf{M}^{-1}\mathbf{N}\,\mathbf{M}) = \operatorname{tr}(\mathbf{M}\,\mathbf{M}^{-1}\mathbf{N}) = \operatorname{tr}(\mathbf{N})$. So the right-hand side is zero. Since $\det \mathbf{M} \neq 0$, this gives $\frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = 0$.

Trace. $\dfrac{\mathrm{d}}{\mathrm{d}t} \operatorname{tr}(\mathbf{M}) = \operatorname{tr}\Big(\dfrac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}\Big) = \operatorname{tr}(\mathbf{MN}) - \operatorname{tr}(\mathbf{NM}) = 0$ by (i).

Trace of the square. The product rule holds for matrices provided we keep the order of the factors:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{M}^2) &= \frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}\mathbf{M} + \mathbf{M}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = (\mathbf{MN} - \mathbf{NM})\mathbf{M} + \mathbf{M}(\mathbf{MN} - \mathbf{NM}) \\ &= \mathbf{M}^2\mathbf{N} - \mathbf{NM}^2, \end{aligned}$$

because the two $\mathbf{MNM}$ terms cancel. This has the same commutator form, now with $\mathbf{M}^2$ in place of $\mathbf{M}$, so its trace is zero by (i). Hence $\operatorname{tr}(\mathbf{M}^2)$ is constant too. All three quantities are independent of t.

> **Ask yourself** How can we solve $\mathrm{d}\mathbf{M}/\mathrm{d}t = \mathbf{MN} - \mathbf{NM}$ when $\mathbf{N}$ depends on t as well?

Split off the part of $\mathbf{N}$ that commutes with everything. Write $\mathbf{N} = t\mathbf{I} + t\mathbf{J}$ with $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. The identity commutes with $\mathbf{M}$, so $\mathbf{MN} - \mathbf{NM} = t(\mathbf{MJ} - \mathbf{JM})$. With $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\mathbf{MJ} - \mathbf{JM} = \begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix} - \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -c & a - d \\ 0 & c \end{pmatrix}.$$

Comparing entries gives four scalar equations:

$$a' = -tc, \qquad b' = t(a-d), \qquad c' = 0, \qquad d' = tc.$$

We solve them in order, using the values A, B, C, D at $t = 0$:

- $c' = 0$, so $c = C$.
- Then $a' = -Ct$ and $d' = Ct$, so $a = A - \frac{1}{2}Ct^2$ and $d = D + \frac{1}{2}Ct^2$.
- Now $a - d = A - D - Ct^2$, so $b' = (A-D)t - Ct^3$, giving $b = B + \frac{1}{2}(A-D)t^2 - \frac{1}{4}Ct^4$.

Therefore

$$\mathbf{M} = \begin{pmatrix} A - \frac{1}{2}Ct^2 & B + \frac{1}{2}(A-D)t^2 - \frac{1}{4}Ct^4 \\ C & D + \frac{1}{2}Ct^2 \end{pmatrix}.$$

As a check, the trace is $A + D$, and expanding $ad - bc$ shows that every power of t cancels, leaving $AD - BC$. This agrees with the first half of the part.

Part (iv)

Now $\mathrm{d}\mathbf{M}/\mathrm{d}t = \mathbf{MN}$, so $\mathbf{M}^{-1}\,\mathrm{d}\mathbf{M}/\mathrm{d}t = \mathbf{N}$, and (ii) gives

$$\operatorname{tr}(\mathbf{N}) = \frac{1}{\det \mathbf{M}} \frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}).$$

So $\operatorname{tr}\mathbf{N} = 0$ for all t exactly when $\det\mathbf{M}$ is constant. A constant trace says nothing about the determinant, so the answer should be *no*. To prove it we need one counterexample.

Ask yourself How do we build a counterexample, when the question gives us $\mathbf{N}$ only through the differential equation?

Work backwards. Choose any non-singular $\mathbf{M}(t)$ with constant non-zero trace and non-constant determinant, and *define* $\mathbf{N} = \mathbf{M}^{-1}\,\mathrm{d}\mathbf{M}/\mathrm{d}t$. Then the equation holds automatically. Take

$$\mathbf{M} = \begin{pmatrix} 1 & t \\ -t & 1 \end{pmatrix}, \qquad \operatorname{tr}\mathbf{M} = 2, \qquad \det\mathbf{M} = 1 + t^2 \neq 0.$$

Then

$$\mathbf{N} = \mathbf{M}^{-1}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \frac{1}{1+t^2}\begin{pmatrix} 1 & -t \\ t & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{1}{1+t^2}\begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix}.$$

Multiplying out confirms that $\mathbf{MN} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathrm{d}\mathbf{M}/\mathrm{d}t$. Also $\mathbf{N}$ is non-singular, since $\det\mathbf{N} = 1/(1+t^2)$. Yet

$$\operatorname{tr}(\mathbf{N}) = \frac{2t}{1+t^2},$$

which is non-zero for every $t \neq 0$. So it is **not** necessarily true that $\operatorname{tr}(\mathbf{N}) = 0$. What *is* forced is $\operatorname{tr}(\mathbf{MN}) = \operatorname{tr}(\mathrm{d}\mathbf{M}/\mathrm{d}t) = 0$, which is not the same thing.

Checks

- In (iii), take $A = 1$, $B = 0$, $C = 2$, $D = 3$ and $t = 1$. The formula gives $\mathbf{M} = \begin{pmatrix} 0 & -\frac{3}{2} \\ 2 & 4 \end{pmatrix}$, which has trace 4 and determinant 3, the same as at $t = 0$. Its square has trace $0 - 3 - 3 + 16 = 10$, and so does the initial matrix: $1 + 0 + 0 + 9 = 10$.
- The formula for $\mathbf{M}$ in (iii) agreed to eight decimal places with a computer integration of the differential equation for random A, B, C, D. The three invariants stayed constant when the same integration was run with a random time-dependent $\mathbf{N}$.
- In (iv), at $t = 1$ we have $\mathbf{N} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ with trace 1, while $\det \mathbf{M} = 1 + t^2$ has logarithmic derivative $2t/(1 + t^2) = 1$ there, as (ii) predicts.

Common slips

- Writing $\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{M}^2) = 2\mathbf{M}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}$: matrices need not commute, so both orders must be kept.
- Assuming $\operatorname{tr}(\mathbf{MN}) = \operatorname{tr}\mathbf{M} \operatorname{tr}\mathbf{N}$. That is false. Only the linear rule and the cyclic rule hold.
- In (iv), arguing from $\operatorname{tr}(\mathbf{MN}) = 0$ straight to $\operatorname{tr}(\mathbf{N}) = 0$. A "no" answer needs an explicit counterexample.

Beyond STEP

Part (ii) is the 2×2 case of Jacobi's formula, which holds for $n \times n$ matrices. Combined with $\mathbf{M}' = \mathbf{MN}$ it gives $\det \mathbf{M}(t) = \det \mathbf{M}(0) \exp \int_0^t \operatorname{tr}\mathbf{N}$. For $\mathbf{N}$ constant this becomes the famous identity $\det e^{\mathbf{N}} = e^{\operatorname{tr}\mathbf{N}}$. Equations of the form $\mathbf{M}' = \mathbf{MN} - \mathbf{NM}$ are called Lax equations. The quantities $\operatorname{tr}(\mathbf{M}^k)$ are always conserved, which is why they appear in the theory of integrable systems. Strang's *Introduction to Linear Algebra* is a good place to read about traces, determinants and matrix exponentials.

WORKED SOLUTION

STEP III 2024, Question 6

Pure • Linear Differential Equations

STEP III 2024, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

The control u is added equally to every equation, so any combination of the coordinates whose coefficients add up to zero doesn't feel u at all. If we choose that combination well, its derivative is a multiple of itself, so it grows or decays exponentially whatever u does. An exponential is never zero unless it starts at zero, which gives the necessary conditions. Once those hold, the motion collapses to a single first-order equation that we can solve with a constant u.

Solution

Part (i)

(a) Subtracting the two equations, the u terms cancel:

$$\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{\mathrm{d}y}{\mathrm{d}t} = (-x + 3y + u) - (x + y + u) = -2(x - y).$$

So $w = x - y$ satisfies $\frac{\mathrm{d}w}{\mathrm{d}t} = -2w$, giving

$$x - y = (x_0 - y_0)\,\mathrm{e}^{-2t}$$

for every function u. If the particle is at the origin at some $t > 0$, then $x - y = 0$ there. Since $\mathrm{e}^{-2t} \neq 0$, this forces $x_0 - y_0 = 0$, that is, $x_0 = y_0$.

(b) With $x_0 = y_0$, the formula in (a) gives $x - y = 0$ for *all* t, so the particle stays on the line $y = x$. Putting $y = x$ into the first equation,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 2x + u.$$

For constant u this is linear with constant coefficients. The particular solution is the constant $-\frac{1}{2}u$, so

$$x = -\tfrac{1}{2}u + \left(x_0 + \tfrac{1}{2}u\right)\mathrm{e}^{2t}.$$

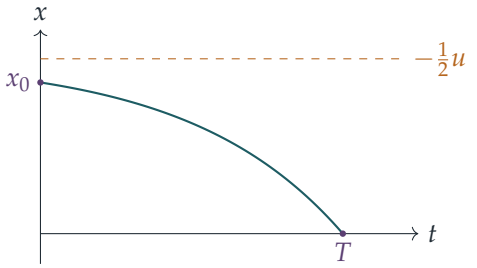
We need $x(T) = 0$ (and then $y(T) = x(T) = 0$ too):

$$\left(x_0 + \tfrac{1}{2}u\right) \mathrm{e}^{2T} = \tfrac{1}{2}u \quad \Longrightarrow \quad x_0 \,\mathrm{e}^{2T} = \tfrac{1}{2}u \left(1 - \mathrm{e}^{2T}\right).$$

Since $T > 0$, $\mathrm{e}^{2T} \neq 1$, so we can divide:

$$\boxed{u = -\frac{2x_0 \,\mathrm{e}^{2T}}{\mathrm{e}^{2T} - 1}}$$

The sign makes sense. Left alone, $\dot{x} = 2x$ pushes the particle away from the origin, so u must push against x_0. (If $x_0 = 0$ the particle is already at rest at the origin with $u = 0$.)



The sketch shows the case $x_0 = 1$, $T = 1$. The particle starts just below the unstable equilibrium $x = -\frac{1}{2}u$ and is driven away from it, reaching the origin exactly at time T.

Part (ii)

(a) Following part (i), we look for $ax + by + cz$ with $a + b + c = 0$ (so u cancels) whose derivative is a multiple of itself. The pattern "x, then $-2y$, then z" suggests $w = x - 2y + z$:

$$\begin{aligned}\frac{\mathrm{d}w}{\mathrm{d}t} &= (4y - 5z + u) - 2(x - 2z + u) + (x - 2y + u) \\ &= -x + 2y - z = -w.\end{aligned}$$

Hence $x - 2y + z = (x_0 - 2y_0 + z_0)\,\mathrm{e}^{-t}$, whatever u is. At the origin $w = 0$, and $\mathrm{e}^{-t} \neq 0$, so $x_0 - 2y_0 + z_0 = 0$. That is, $y_0 = \frac{1}{2}(x_0 + z_0)$, the mean of x_0 and z_0.

? **Ask yourself** Is there a second combination that also ignores u?

The last two equations both contain $x + u$, so subtracting them removes x as well as u. Let $v = y - z$. Then

$$\frac{\mathrm{d}v}{\mathrm{d}t} = (x - 2z + u) - (x - 2y + u) = 2(y - z) = 2v,$$

so $y - z = (y_0 - z_0)\,\mathrm{e}^{2t}$.

(b) If the particle is at the origin at some $t > 0$, then $v = 0$ there. Since $\mathrm{e}^{2t} \neq 0$, we get $y_0 = z_0$. Combining this with (a), $x_0 = 2y_0 - z_0 = y_0$, so $x_0 = y_0 = z_0$.

(c) With $x_0 = y_0 = z_0$, both w and v start at 0, so they stay 0 for all t: $y = z$, and then $x = 2y - z = y$. The particle moves along the line $x = y = z$, and each equation becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 4x - 5x + u = -x + u$$

(the second and third give $x - 2x + u$, the same). For constant u,

$$x = u + (x_0 - u)\,\mathrm{e}^{-t}.$$

Setting $x(T) = 0$ gives $u\left(1 - \mathrm{e}^{-T}\right) = -x_0\,\mathrm{e}^{-T}$. Since $T > 0$ we may divide, and multiplying top and bottom by e^{T},

$$\boxed{u = -\frac{x_0}{\mathrm{e}^{T} - 1}}$$

Here the uncontrolled motion $\dot{x} = -x$ already decays towards the origin, but only as $t \to \infty$. A small push towards the origin, which tends to zero as $T \to \infty$, gets the particle there in finite time.

Checks

- Limits: as $T \to \infty$ the answer to (ii)(c) tends to 0, matching the natural decay, while the answer to (i)(b) tends to $-2x_0$, which makes $x_0 = -\frac{1}{2}u$ an equilibrium. The particle then sits there and never arrives, which is why a finite T needs u slightly larger in size. As $T \to 0^{+}$ both answers blow up, as they should.
- Numerical: with $x_0 = 1$, $T = 1$, part (i)(b) gives $u \approx -2.3130$, and stepping the equations forward numerically lands on $(0, 0)$ at $t = 1$ to within 10^{-8}. The same holds for (ii)(c) with $u = -1/(\mathrm{e} - 1) \approx -0.5820$. The invariants $x - y$, $x - 2y + z$ and $y - z$ were also confirmed to follow their exponentials with u a non-constant function of time.

Common slips

- Assuming u is constant in (i)(a) or (ii)(a),(b). Those parts are for *any* u, and the cancellation of u is the whole point.
- In (i)(b) and (ii)(c), forgetting to say *why* $y = x$ (or $x = y = z$) for all time. It follows from the exponential formulas with zero initial value.

Beyond STEP

This is a first taste of *control theory*. Writing the system as $\dot{\mathbf{r}} = A\mathbf{r} + \mathbf{b}u$, the combinations we found are left eigenvectors of A that are orthogonal to $\mathbf{b}$, so no control can affect them. Kalman's rank condition decides whether every state can be steered to the origin. Here it fails, which is exactly why only special starting points work. Strang's *Introduction to Linear Algebra* covers the eigenvector side, and Boyce and DiPrima's *Elementary Differential Equations* covers linear systems of ODEs.

WORKED SOLUTION

STEP III 2024, Question 7

Pure • Irrationality of e Squared

STEP III 2024, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.4, *e Is Irrational, and So Is e^2*: https://beyondstep.keithmcnulty.org/ch01/e-irrational.html

The key idea

Both series are the tails of the exponential series once it has been multiplied by a factorial. Multiply $\mathrm{e} = \sum 1/r!$ by $(2n)!$ and every term up to $r = 2n$ becomes an integer, while everything after is exactly $\mathrm{f}(2n)$; the same happens for $1/\mathrm{e}$ and $\mathrm{g}(2n)$. If e^2 were rational, a combination of these small tails would be forced to be an integer, yet it is squeezed strictly between 0 and 1.

Solution

Write

$$a_k = \frac{1}{(n+1)(n+2)\cdots(n+k)}, \qquad k \geq 1,$$

so that $\mathrm{f}(n) = a_1 + a_2 + a_3 + \cdots$ and $\mathrm{g}(n) = a_1 - a_2 + a_3 - \cdots$. Note that $a_{k+1} = a_k/(n+k+1)$.

Part (i)

Every a_k is positive, so $\mathrm{f}(n) > 0$. Each factor in the denominator of a_k is at least $n+1$, so $a_k \leq 1/(n+1)^k$, with strict inequality once $k \geq 2$ (the factor $n+2$ appears). Comparing with a geometric series of ratio $\frac{1}{n+1} < 1$,

$$\mathrm{f}(n) < \sum_{k=1}^{\infty} \frac{1}{(n+1)^k} = \frac{\frac{1}{n+1}}{1 - \frac{1}{n+1}} = \frac{1}{n}.$$

Hence $0 < \mathrm{f}(n) < \frac{1}{n}$.

Part (ii)

Consecutive terms strictly decrease: $a_{k+1} = a_k/(n+k+1) < a_k$, since $n+k+1 > 1$. Grouping in pairs from the start,

$$\mathrm{g}(n) = (a_1 - a_2) + (a_3 - a_4) + \cdots > 0,$$

as each bracket is positive. Grouping instead after the first term,

$$\mathrm{g}(n) = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots < a_1 = \frac{1}{n+1},$$

as each bracket is again positive. So $0 < \mathrm{g}(n) < \frac{1}{n+1}$.

Part (iii)

Let $m = 2n$. Using $\mathrm{e} = \sum_{r\geq 0} 1/r!$,

$$m!\,\mathrm{e} = \sum_{r=0}^{m} \frac{m!}{r!} + \sum_{r=m+1}^{\infty} \frac{m!}{r!}.$$

For $r \leq m$, $m!/r! = (r+1)(r+2)\cdots m$ is an integer, so the first sum is an integer; call it A. In the second sum, $m!/(m+k)! = 1/\big((m+1)\cdots(m+k)\big)$, which is the kth term of $\mathrm{f}(m)$. So $m!\,\mathrm{e} = A + \mathrm{f}(m)$, and $(2n)!\,\mathrm{e} - \mathrm{f}(2n) = A$ is an integer.

Similarly $1/\mathrm{e} = \sum_{r\geq 0}(-1)^r/r!$ gives

$$\frac{m!}{\mathrm{e}} = \underbrace{\sum_{r=0}^{m} (-1)^r \frac{m!}{r!}}_{B\in\mathbb{Z}} + \sum_{k=1}^{\infty} \frac{(-1)^{m+k}}{(m+1)(m+2)\cdots(m+k)}.$$

? **Ask yourself** Why does the question use $2n$ rather than n?

Because $m = 2n$ is even, $(-1)^{m+k} = (-1)^k$, so the tail is $-a_1 + a_2 - a_3 + \cdots = -\mathrm{g}(m)$ (with a_k now formed from m). Thus $m!/\mathrm{e} = B - \mathrm{g}(m)$, and $\dfrac{(2n)!}{\mathrm{e}} + \mathrm{g}(2n) = B$ is an integer. (For odd m the tail would be $+\mathrm{g}(m)$ and we would need a minus sign instead.)

Part (iv)

Using the integers A and B from part (iii),

$$\begin{aligned} q\,\mathrm{f}(2n) + p\,\mathrm{g}(2n) &= q\big((2n)!\,\mathrm{e} - A\big) + p\Big(B - \frac{(2n)!}{\mathrm{e}}\Big) \\ &= (2n)!\Big(q\mathrm{e} - \frac{p}{\mathrm{e}}\Big) - qA + pB \\ &= pB - qA, \end{aligned}$$

since $q\mathrm{e} = p/\mathrm{e}$. As p, q, A, B are integers, so is $q\,\mathrm{f}(2n) + p\,\mathrm{g}(2n)$, for every positive integer n.

Part (v)

Suppose, for contradiction, that e^2 is rational. Since $e^2 > 0$ we can write $e^2 = p/q$ with p, q positive integers, and dividing by e gives $qe = p/e$. By part (iv), $N_n = q\,f(2n) + p\,g(2n)$ is an integer for every $n \geq 1$.

? **Ask yourself** How can we make N_n fail to be an integer?

Squeeze it with parts (i) and (ii), applied at $2n$. Since $p, q > 0$,

$$0 < N_n < \frac{q}{2n} + \frac{p}{2n+1} < \frac{p+q}{2n}.$$

Now choose n large enough that $2n > p + q$. Then $0 < N_n < 1$, so N_n is an integer strictly between 0 and 1, which is impossible. Hence e^2 is irrational.

Checks

Small cases can be summed exactly. For $n = 2$, part (iii) says $24e - f(4)$ is an integer; indeed $\sum_{r=0}^{4} 24/r! = 65$, and $f(4) = 24e - 65 \approx 0.2388$, comfortably inside $(0, \frac{1}{4})$. Likewise $f(2) = 2e - 5 \approx 0.4366 < \frac{1}{2}$ and $g(2) = 1 - 2/e \approx 0.2642 < \frac{1}{3}$. A computer check confirmed the bounds of (i) and (ii) for $n \leq 24$, the integrality in (iii) for $2n \leq 22$, and that $0 < q\,f(2n) + p\,g(2n) < (p+q)/(2n)$ for random p, q.

Common slips

- Claiming $a_k < 1/(n+1)^k$ for all k: equality holds at $k = 1$, so the strict inequality comes from $k \geq 2$.
- Forgetting the sign of the tail in the $1/e$ expansion: it is $-g(2n)$ only because $2n$ is even.
- Stopping at "N_n is small": the contradiction needs $N_n > 0$ as well, which is why the lower bounds in (i) and (ii) matter.

Beyond STEP

This is Fourier's classic proof that e is irrational, extended by splitting $e^2 = p/q$ into $qe = p/e$ so that both series converge fast. The same idea, with more care, shows e^r is irrational for every non-zero rational r, and Liouville used it to show that e is not a root of any quadratic with integer coefficients. Bounding tails of series is a core technique of first-year analysis; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2024, Question 8

Pure • Intersecting Conics

STEP III 2024, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

A quadratic expression of the form $y^2 - x^2 + \dots$ splits into two lines of gradients ± 1 exactly when completing the square leaves no constant behind. The trick in the rest of the question is that adding a multiple of one curve's equation to the other's doesn't lose any common points, and a well-chosen multiple turns the messy quartic problem into "which lines meet a parabola", which is just two quadratics.

Solution

Part (i)

A product is zero exactly when one of its factors is zero, so the points satisfying $(y - x + 3)(y + x - 5) = 0$ are those on $y = x - 3$ (gradient 1) together with those on $y = -x + 5$ (gradient -1).

For the general equation, complete the square in both variables:

$$y^2 - x^2 + py + qx + r = \left(y + \tfrac{1}{2}p\right)^2 - \left(x - \tfrac{1}{2}q\right)^2 - K, \qquad \text{where } K = \tfrac{1}{4}(p^2 - q^2) - r.$$

Write $Y = y + \frac{1}{2}p$ and $X = x - \frac{1}{2}q$, a translation, which doesn't change gradients. The equation becomes $(Y - X)(Y + X) = K$.

If $p^2 - q^2 = 4r$, then $K = 0$ and the equation is $(Y - X)(Y + X) = 0$, which by the first paragraph is the pair of lines $Y = X$ and $Y = -X$, with gradients 1 and -1. In the original coordinates these are

$$y - x + \tfrac{1}{2}(p + q) = 0 \quad \text{and} \quad y + x + \tfrac{1}{2}(p - q) = 0.$$

Conversely, suppose the equation represents a pair of lines with gradients ± 1; we show $K = 0$. One of the lines has gradient 1, so has the form $Y = X + c$, and every point of it satisfies the equation. On it, $(Y - X)(Y + X) = c(2X + c)$, so $2cX + c^2 = K$ for *every* real X. A linear function of X is constant only if its coefficient vanishes, so $c = 0$, and then $K = c^2 = 0$. Hence $p^2 - q^2 = 4r$, and the "if and only if" is proved.

Part (ii)

If a point lies on C_1, then $y^2 + 2sy + s(s+1) - x = 0$ there; if it lies on C_2, then $y - x^2 = 0$ there. So at a common point the expression

$$y^2 + 2sy + s(s+1) - x + k(y - x^2)$$

is $0 + k \cdot 0 = 0$, whatever the value of k.

With $k = 1$ the equation is

$$y^2 - x^2 + (2s+1)y - x + s(s+1) = 0,$$

which has the form of part (i) with $p = 2s+1$, $q = -1$, $r = s(s+1)$. Then

$$p^2 - q^2 = 4s^2 + 4s + 1 - 1 = 4s(s+1) = 4r,$$

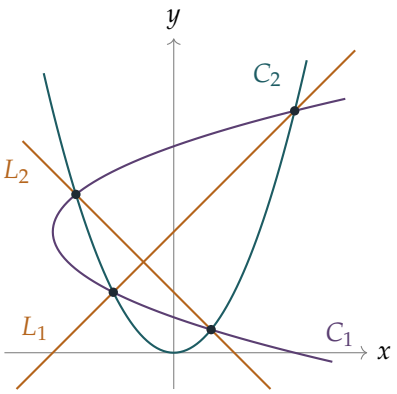
so by part (i) this is a pair of straight lines with gradients 1 and -1. Using the factorisation from part (i), with $\frac{1}{2}(p+q) = s$ and $\frac{1}{2}(p-q) = s+1$, the equation is

$$(y - x + s)(y + x + s + 1) = 0,$$

so the lines are

$$L_1 : \ y = x - s \qquad \text{and} \qquad L_2 : \ y = -x - s - 1.$$

By the first paragraph every point of $C_1 \cap C_2$ satisfies this equation, so all four intersection points lie on $L_1 \cup L_2$. (This holds for every s; we never needed the four points to exist.)



The figure shows the case $s = -2$: C_1 is $x = (y-2)^2 - 2$, and the four common points of C_1 and C_2 lie two on each of the lines $y = x + 2$ and $y = 1 - x$.

Part (iii)

Every common point of C_1 and C_2 lies on C_2 and on L_1 or L_2. Substituting $y = x^2$:

$$\begin{aligned} &\text{on } L_1 : \quad x^2 - x + s = 0, \\ &\text{on } L_2 : \quad x^2 + x + s + 1 = 0. \end{aligned}$$

A point on C_2 is fixed by its x-coordinate, so each line meets C_2 in at most two points. Four distinct common points therefore force each quadratic to have two distinct real roots, so both discriminants are positive:

$$1 - 4s > 0 \quad \text{and} \quad 1 - 4(s+1) > 0.$$

The second gives $s < -\frac{3}{4}$, as required. (The first, $s < \frac{1}{4}$, is then automatic.)

Ask yourself In part (iv), why isn't it enough to find four points on C_2 and $L_1 \cup L_2$?

Part (ii) only showed that common points of C_1 and C_2 lie on the lines; we need the reverse direction too, and we must make sure the four points don't coincide.

Part (iv)

Suppose $s < -\frac{3}{4}$. Then both discriminants above are positive ($-3 - 4s > 0$, and $1 - 4s > 4 > 0$), so L_1 meets C_2 at two distinct points and so does L_2.

These four points are distinct. A point counted twice would lie on both L_1 and L_2. The lines meet where $x - s = -x - s - 1$, that is, at $\left(-\frac{1}{2}, -\frac{1}{2} - s\right)$. This lies on C_2 only if $-\frac{1}{2} - s = \frac{1}{4}$, that is, $s = -\frac{3}{4}$, which is excluded. So we have four distinct points.

Each of them lies on C_1. At each point $y - x^2 = 0$ (it is on C_2), and the $k = 1$ equation of part (ii) holds (it is on L_1 or L_2). Subtracting,

$$y^2 + 2sy + s(s+1) - x = 0,$$

which is the equation of C_1.

So C_1 and C_2 have at least four distinct common points; by part (iii)'s argument they can't have more than four. Hence they intersect at exactly four distinct points.

Checks

- Substituting $y = x^2$ into C_1 gives the quartic $x^4 + 2sx^2 - x + s^2 + s = 0$. Multiplying out $(x^2 - x + s)(x^2 + x + s + 1)$ gives exactly this, which confirms the pair of lines.
- With $s = -2$ the four points are $(2, 4)$, $(-1, 1)$ and $\left(\frac{1}{2}(-1 \pm \sqrt{5}), \frac{1}{2}(3 \mp \sqrt{5})\right)$; each satisfies $x = (y-2)^2 - 2$, e.g. $(2,4)$: $(4-2)^2 - 2 = 2$.
- At the boundary $s = -\frac{3}{4}$ the quadratics are $x^2 - x - \frac{3}{4} = 0$ (roots $\frac{3}{2}, -\frac{1}{2}$) and $\left(x + \frac{1}{2}\right)^2 = 0$, so there are only two distinct common points, with $x = -\frac{1}{2}$ shared by both lines. A computer count of real roots of the quartic gave 4 for $s = -0.76, -1, -2$ and 2 for $s = -0.74, -0.5$.

Common slips

- Proving only the "if" direction in part (i): a pair of lines must give $p^2 - q^2 = 4r$, and you need an argument (such as the one above) for that.
- In part (iv), forgetting to check that the two lines don't share a point of C_2, or forgetting to show that the points found actually lie on C_1.

Beyond STEP

The family $F + kG = 0$ is a *pencil of conics*: every member passes through the common points of $F = 0$ and $G = 0$. Choosing k so that the member is degenerate (a pair of lines) is a classical way to find the intersections of two conics. In projective geometry this is done by making a 3×3 determinant vanish, which gives a cubic in k. Here the pencil happened to be degenerate for $k = 1$ whatever s is. Strang's *Introduction to Linear Algebra* covers the symmetric matrices behind conics. Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

WORKED SOLUTION

STEP III 2024, Question 9

Mechanics • Oblique Collisions

STEP III 2024, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything happens in the plane of the table, so look down from above: each sphere becomes a circle of radius r centred at its lowest point. For smooth spheres the impulse acts along the line of centres, so we split velocities into the component along that line (where momentum and restitution apply) and the component across it (which is unchanged). After that, B moves in a straight line, and whether it hits C is a question about the distance from a point to a line.

Solution

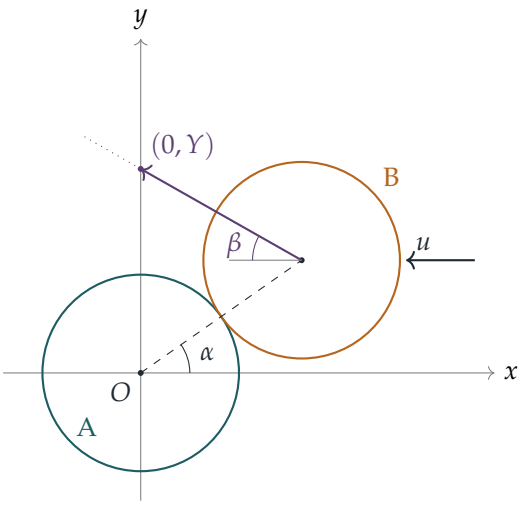
We model the spheres as smooth and equal in mass, on a smooth table, so there is no friction anywhere and each sphere moves in a straight line at constant speed except during the impact. We work with the lowest points, which move exactly as the centres do horizontally; two spheres of radius r touch when their lowest points are $2r$ apart.

Part (i)

B's lowest point moves along the line $y = 2r\sin\alpha$. It touches A when its distance from O is $2r$, that is at the point $(2r\cos\alpha,\, 2r\sin\alpha)$ (taking the positive x-coordinate, since B comes from large positive x). So at impact the line of centres, from A to B, has unit vector

$$\mathbf{n} = \begin{pmatrix}\cos\alpha\\ \sin\alpha\end{pmatrix}, \qquad \text{and} \qquad \mathbf{t} = \begin{pmatrix}-\sin\alpha\\ \cos\alpha\end{pmatrix}$$

is perpendicular to it.



Before impact B has velocity $\begin{pmatrix} -u \\ 0 \end{pmatrix}$, with components $-u\cos\alpha$ along **n** and $u\sin\alpha$ along **t**. A is at rest.

The spheres are smooth, so the impulse between them is along **n**. Hence B's **t**-component stays $u\sin\alpha$ (and A gains no **t**-component). Let the **n**-components after impact be v_A and v_B. Conservation of momentum along **n** (equal masses) and Newton's law of restitution give

$$\begin{aligned} v_A + v_B &= -u\cos\alpha, \\ v_B - v_A &= \tfrac{1}{3}\big(0 - (-u\cos\alpha)\big) = \tfrac{1}{3}u\cos\alpha. \end{aligned}$$

(The approach speed along **n** is $u\cos\alpha$, and B, on the positive-**n** side, separates from A.) Adding, $v_B = -\frac{1}{3}u\cos\alpha$, and $v_A = -\frac{2}{3}u\cos\alpha$.

So B's velocity after impact is

$$\begin{aligned} -\tfrac{1}{3}u\cos\alpha\begin{pmatrix}\cos\alpha\\ \sin\alpha\end{pmatrix} + u\sin\alpha\begin{pmatrix}-\sin\alpha\\ \cos\alpha\end{pmatrix} &= \begin{pmatrix} -\frac{1}{3}u(\cos^2\alpha + 3\sin^2\alpha) \\ \frac{2}{3}u\sin\alpha\cos\alpha \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{3}u(1 + 2\sin^2\alpha) \\ \frac{2}{3}u\sin\alpha\cos\alpha \end{pmatrix}, \end{aligned}$$

using $\cos^2\alpha = 1 - \sin^2\alpha$. This is the required result.

Part (ii)

Both components are nonzero, with the x-component negative and the y-component positive (as α is acute). So B now travels up and to the left, and the gradient of its path has size

$$\frac{\frac{2}{3}u\sin\alpha\cos\alpha}{\frac{1}{3}u(1+2\sin^2\alpha)} = \frac{2\sin\alpha\cos\alpha}{1+2\sin^2\alpha} = \tan\beta,$$

where β is the acute angle the path makes with the negative x-direction. B's lowest point starts at $(2r\cos\alpha, 2r\sin\alpha)$. To reach the y-axis it must decrease x by $2r\cos\alpha$, and meanwhile y increases by $2r\cos\alpha\tan\beta$. So it crosses the y-axis at $(0, Y)$ with

$$Y = 2r\sin\alpha + 2r\cos\alpha\tan\beta = 2r(\cos\alpha\tan\beta + \sin\alpha),$$

as required.

Part (iii)

After the impact B moves in a straight line at constant speed, along the line ℓ through $(0, Y)$ with unit direction $\mathbf{d} = (-\cos\beta, \sin\beta)$. C stays at rest at $(0, h)$ until something hits it. B strikes C exactly when the distance from $(0, h)$ to B's path drops below $2r$.

The perpendicular distance from $(0, h)$ to ℓ is the size of the component of $(0, h) - (0, Y) = (0, h - Y)$ perpendicular to $\mathbf{d}$. A unit normal to ℓ is $(\sin\beta, \cos\beta)$, so this distance is

$$\big|(0,\, h - Y)\cdot(\sin\beta, \cos\beta)\big| = |h - Y|\cos\beta,$$

since β is acute and so $\cos\beta > 0$. If $h > Y + 2r\sec\beta$, then $h - Y > 2r\sec\beta > 0$ and the distance is $(h - Y)\cos\beta > 2r$. Every point of B's path, not just the closest point of the whole line, is at least this far from C's lowest point. So B never comes within $2r$ of C and does not strike it.

(Nothing else gets in the way: A moves off with velocity $-\frac{2}{3}u\cos\alpha\,\mathbf{n}$, into $y < 0$, away from both B and C.)

> **Ask yourself** Part (iv) asks us to prove $Y < 2r\sec\beta$. Can we turn it into a single trigonometric fact?

Yes: multiply through by $\cos\beta > 0$ and the left-hand side becomes a compound-angle formula.

Part (iv)

Dividing by $2r$ and multiplying by $\cos\beta > 0$, the inequality $Y < 2r\sec\beta$ is equivalent to

$$\cos\alpha\sin\beta + \sin\alpha\cos\beta < 1, \qquad \text{that is,} \qquad \sin(\alpha + \beta) < 1.$$

Certainly $\sin(\alpha + \beta) \le 1$, with equality only if $\alpha + \beta = \frac{\pi}{2}$ (as $0 < \alpha + \beta < \pi$). That would mean $\tan\beta = \cot\alpha$, i.e.

$$\frac{2\sin\alpha\cos\alpha}{1 + 2\sin^2\alpha} = \frac{\cos\alpha}{\sin\alpha} \iff 2\sin^2\alpha = 1 + 2\sin^2\alpha$$

(dividing by $\cos\alpha \neq 0$), which is impossible. Hence $\sin(\alpha + \beta) < 1$, and so $Y < 2r\sec\beta$.

Hence. Now

$$Y + 2r\sec\beta < 4r\sec\beta,$$

so we need an upper bound on $\sec\beta$ over all acute α, i.e. an upper bound on $\tan\beta$. Dividing top and bottom by $\cos^2\alpha$ and writing $t = \tan\alpha > 0$,

$$\tan\beta = \frac{2\tan\alpha}{\sec^2\alpha + 2\tan^2\alpha} = \frac{2t}{1+3t^2} \le \frac{2t}{2\sqrt{3}\,t} = \frac{1}{\sqrt{3}},$$

by the AM–GM inequality $1 + 3t^2 \ge 2\sqrt{3}\,t$ (equality when $t = 1/\sqrt{3}$, i.e. $\alpha = \frac{\pi}{6}$). So $0 < \beta \le \frac{\pi}{6}$, which gives $\sec\beta \le \sec\frac{\pi}{6} = \frac{2}{\sqrt{3}}$. Therefore, for every acute α,

$$Y + 2r\sec\beta < 4r\sec\beta \le \frac{8r}{\sqrt{3}}.$$

If $h > \dfrac{8r}{\sqrt{3}}$ then $h > Y + 2r\sec\beta$, and by part (iii) B does not strike C, whatever the value of α.

Checks

- Momentum: A's velocity $-\frac{2}{3}u\cos\alpha\,(\cos\alpha, \sin\alpha)$ plus B's velocity from part (i) adds up to $(-u, 0)$, as it should for equal masses.
- When $\alpha = 0$ (a head-on impact) the formula gives B velocity $(-\frac{1}{3}u, 0)$ and A gets $(-\frac{2}{3}u, 0)$: the usual one-dimensional answer for equal masses with $e = \frac{1}{3}$.
- With $\alpha = \frac{\pi}{6}$ we get $\tan\beta = 1/\sqrt{3}$, so $\beta = \frac{\pi}{6}$, $\alpha + \beta = \frac{\pi}{3}$ and $Y = 2r(\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}} + \frac{1}{2}) = 2r$, comfortably less than $2r\sec\beta = 4r/\sqrt{3} \approx 2.31r$. Then $Y + 2r\sec\beta \approx 4.31r$, below $8r/\sqrt{3} \approx 4.62r$.
- We confirmed by computer, for 20,000 random values of α, that stepping B's straight path gives the crossing point Y above, that B misses C just above $h = Y + 2r\sec\beta$ and hits it just below, and that the largest value of $\sec\beta$ is $1.1547\ldots = 2/\sqrt{3}$.

Common slips

- Applying restitution to the x-components instead of the components along the line of centres. The impulse between smooth spheres acts along the line of centres, which here is at angle α to the x-axis.
- Taking the line of centres at impact to be along the x-axis or the y-axis. The contact point is fixed by the spheres being $2r$ apart.
- In part (iii), testing whether C's lowest point lies within $2r$ of the single point $(0, Y)$ rather than of the whole path.

Beyond STEP

Splitting a velocity into components along and across a chosen unit vector is projection onto an orthonormal basis, an idea that runs through first-year linear algebra and

mechanics. Oblique collisions of smooth spheres are a first model of scattering, which physicists use to study gases and particle collisions. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions in two dimensions with this vector approach.

WORKED SOLUTION

STEP III 2024, Question 10

Mechanics • Toppling and Sliding

STEP III 2024, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

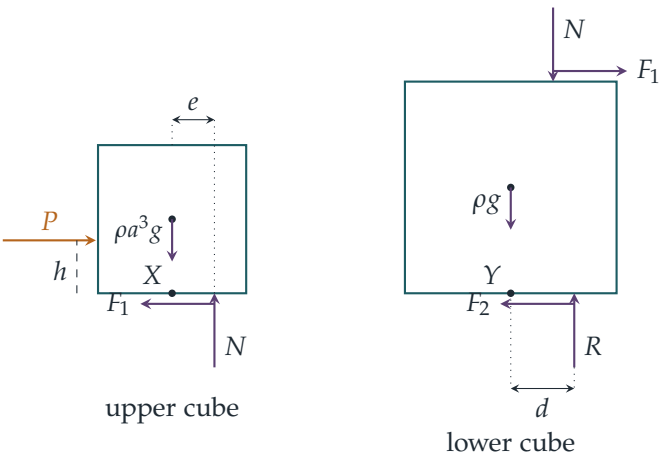
www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every part is a comparison between four ways in which equilibrium can fail: the upper cube slides, the lower cube slides, the upper cube topples, or the two cubes topple together. Each of these happens at its own critical value of P, and whichever critical value is smallest wins as P grows from zero. Taking moments about the right point, Y for the whole system and X for the upper cube alone, gives the positions of the normal reactions in part (i) with very little work, and toppling begins when one of these reactions reaches the edge of the face it acts on.

Solution

We model the cubes as rigid bodies. The lower cube has mass ρ and the upper cube has mass ρa^3. Take P to act horizontally to the right, at height h above the top face of the lower cube, so at height $1 + h$ above the ground. Since P is horizontal, its moment about any point depends only on its height, not on where along the face it acts.



Here N and F_1 are the normal reaction and friction between the cubes (shown on each cube by Newton's third law), acting at distance e from X; R and F_2 are the normal reaction and friction from the ground, acting at distance d from Y.

Part (i)

Whole system. The only external forces are P, the two weights, R and F_2. Resolving,

$$F_2 = P, \qquad R = \rho g + \rho a^3 g = (1 + a^3)\rho g.$$

Take moments about Y. The weights both act along the vertical line through Y (each centre of mass is directly above Y), and F_2 acts along the ground through Y, so none of these has a moment. That leaves P, at height $1 + h$, balanced by R at distance d:

$$R\,d = P(1 + h) \quad \Longrightarrow \quad d = \frac{P(1 + h)}{(1 + a^3)\rho g},$$

measured from Y in the direction of P. This is the required result.

Upper cube alone. Resolving gives $F_1 = P$ and $N = \rho a^3 g$. Taking moments about X, the weight and F_1 have no moment, so $Ne = Ph$ and

$$e = \frac{Ph}{\rho a^3 g},$$

again measured from X in the direction of P.

Part (ii)

While nothing has moved, the friction forces are $F_1 = P$ between the cubes and $F_2 = P$ at the ground. Slipping happens at a contact once the friction needed there exceeds μ times the normal reaction, so

$$\text{upper cube slips on lower when } P > \mu\rho a^3 g,$$
$$\text{lower cube slips on ground when } P > \mu(1 + a^3)\rho g.$$

Since $a^3 < 1 + a^3$, the first threshold is always strictly smaller. As P increases from zero (and if no toppling intervenes), the upper cube reaches its limit first and starts to slide while the ground friction is still well within its limit. Once the upper cube is sliding, the friction it exerts on the lower cube is $\mu\rho a^3 g$, still less than the $\mu(1 + a^3)\rho g$ that the ground can supply, so the lower cube stays put. So equilibrium is broken by the upper cube slipping, not by the lower cube slipping.

Part (iii)

A body on a flat face is about to topple when the normal reaction on it reaches the edge of that face. The upper cube's base has half-width $\frac{1}{2}a$, and the lower cube's base has

half-width $\frac{1}{2}$. So, measuring every force in units of ρg, the four critical values of P are

$$P_{\text{slip}} = \mu a^3 \text{ (upper on lower)}, \qquad P_{\text{ground}} = \mu(1+a^3),$$
$$P_{\text{upper}} = \frac{a^4}{2h} \text{ (from } e = \tfrac{1}{2}a\text{)}, \qquad P_{\text{both}} = \frac{1+a^3}{2(1+h)} \text{ (from } d = \tfrac{1}{2}\text{)}.$$

Toppling of the lower cube carries the upper one with it, which is why $d = \frac{1}{2}$ means the two cubes topple together.

Now put $a = 1$, so $P_{\text{slip}} = \mu$, $P_{\text{upper}} = \frac{1}{2h}$ and $P_{\text{both}} = \frac{1}{1+h}$. By part (ii) we can ignore P_{ground}.

? **Ask yourself** Do we need to worry about the upper cube toppling on its own?

No, but it must be checked. Since $h < a = 1$ we have $2h < 1 + h$, so

$$P_{\text{upper}} = \frac{1}{2h} > \frac{1}{1+h} = P_{\text{both}}.$$

The two cubes would topple together before the upper cube could topple alone. So the contest is only between $P_{\text{slip}} = \mu$ and $P_{\text{both}} = 1/(1+h)$:

- if $\mu(1+h) < 1$, then $\mu < \frac{1}{1+h}$, the slipping threshold is reached first, and the upper cube slips on the lower one;
- if $\mu(1+h) > 1$, then $\frac{1}{1+h} < \mu$, the toppling threshold is reached first, and the two cubes topple together.

Part (iv)

If no slipping occurs, the contest is between P_{upper} and P_{both}. The upper cube topples first exactly when

$$\frac{a^4}{2h} < \frac{1+a^3}{2(1+h)} \iff a^4(1+h) < h(1+a^3) \iff a^4 < h\big(1+a^3-a^4\big) = h\big(1+a^3(1-a)\big),$$

where we multiplied through by the positive quantity $2h(1+h)$. This is precisely the given condition, so under it equilibrium is broken by the upper cube toppling.

Part (v)

With $a = \frac{1}{2}$ we need three things.

- The condition of (iv): $h\big(1 + \frac{1}{8}\cdot\frac{1}{2}\big) > \frac{1}{16}$, that is, $\frac{17}{16}h > \frac{1}{16}$, so $h > \frac{1}{17}$.
- The setup requires $h < a = \frac{1}{2}$.
- No slipping before toppling: $P_{\text{upper}} < P_{\text{slip}}$, that is, $\dfrac{1/16}{2h} < \dfrac{\mu}{8}$, or $\mu > \dfrac{1}{4h}$. (Slipping on the ground is never first, by (ii).) Since we also need $\mu < 1$, this requires $h > \frac{1}{4}$.

So choose $h = \frac{2}{5}$, which satisfies $\frac{1}{4} < h < \frac{1}{2}$ and $h > \frac{1}{17}$. Then we need $\frac{5}{8} < \mu < 1$; take $\mu = \frac{3}{4}$. The critical values (in units of ρg) are then

$$P_{\text{upper}} = \tfrac{5}{64} \approx 0.078, \quad P_{\text{slip}} = \tfrac{3}{32} \approx 0.094, \quad P_{\text{both}} = \tfrac{45}{112} \approx 0.402, \quad P_{\text{ground}} = \tfrac{27}{32} \approx 0.844.$$

The smallest is P_{upper}, so as P grows the upper cube topples before anything slips: the situation of (iv) does occur.

Checks

- As $P \to 0$, both d and e tend to 0: with no push, each reaction acts directly below the centre of mass, as it should.
- Taking moments for the lower cube alone (its weight, R and F_2 at distance d, and N and F_1 reversed at distance e from X) gives $Rd = Ne + F_1 \cdot 1$, which is $P(1+h) = Ph + P$. So the two answers in (i) are consistent.
- For $a = \frac{1}{2}$, $h = \frac{2}{5}$, $\mu = \frac{3}{4}$ the four thresholds above were computed exactly, and the comparisons in (ii)–(iv) were confirmed by computer on many random choices of a, h and μ.

Common slips

- In (iii), comparing only slipping with toppling of both cubes, and never showing that the upper cube cannot topple first.
- Taking moments about a point on the ground for the upper cube, which brings in the friction F_1 at the interface; moments about X avoid it.
- Using h instead of $1 + h$ as the height of P above the ground in the whole-system moment equation.

Beyond STEP

Each failure mode here is an inequality on the forces, and equilibrium survives as long as all of them hold at once; the first one to fail is the one with the smallest threshold. This idea of a set of linear constraints with a "binding" one is the heart of the limit analysis used for masonry and stacked structures in engineering, and of linear programming more widely. For a careful treatment of friction, torque and the statics of rigid bodies, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2024, Question 11

Probability and Statistics • Binomial Distribution Expectation

STEP III 2024, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

The identity in (i) pairs each term $r\binom{2n}{r}$ in the bottom half of the sum with a term in the top half. That symmetry is exactly what X needs: the larger of heads and tails is either the number of heads or the number of tails, and swapping the roles of heads and tails folds the expectation onto the top half of the sum. After that, (iii) is a ratio test, and (iv) needs only a line.

Solution

Part (i)

For $1 \leq r \leq 2n$, cancelling r against $r!$ gives

$$r\binom{2n}{r} = \frac{r\,(2n)!}{r!\,(2n-r)!} = \frac{(2n)!}{(r-1)!\,(2n-r)!}.$$

On the other side, $2n-(2n+1-r) = r-1$, so

$$(2n+1-r)\binom{2n}{2n+1-r} = \frac{(2n+1-r)\,(2n)!}{(2n+1-r)!\,(r-1)!} = \frac{(2n)!}{(2n-r)!\,(r-1)!},$$

again cancelling the top factor against the factorial. (We need $1 \leq r \leq 2n$ so that $2n+1-r$ lies between 1 and $2n$, which makes both factorials sensible.) The two expressions agree, so the identity holds.

For the sum, the $r = 0$ term is zero. Split the rest into $1 \leq r \leq n$ and $n+1 \leq r \leq 2n$. In the lower block, put $s = 2n+1-r$. As r runs from 1 to n, s runs from $2n$ down to $n+1$, and by the identity

$$\sum_{r=1}^{n} r\binom{2n}{r} = \sum_{r=1}^{n}(2n+1-r)\binom{2n}{2n+1-r} = \sum_{s=n+1}^{2n} s\binom{2n}{s}.$$

So the lower block equals the upper block, and

$$\sum_{r=0}^{2n} r\binom{2n}{r} = 2\sum_{r=n+1}^{2n} r\binom{2n}{r},$$

as required.

Part (ii)

Let H be the number of heads, so $H \sim \mathrm{B}(2n, \frac{1}{2})$ and $\mathrm{P}(H = r) = \binom{2n}{r}/2^{2n}$. The number of tails is $2n - H$, so $X = \max(H, 2n - H)$. That is, $X = H$ when $H \geq n+1$, $X = n$ when $H = n$, and $X = 2n - H$ when $H \leq n-1$. Hence

$$2^{2n}\,\mathrm{E}(X) = \sum_{r=n+1}^{2n} r\binom{2n}{r} + n\binom{2n}{n} + \sum_{r=0}^{n-1}(2n-r)\binom{2n}{r}.$$

In the last sum put $s = 2n - r$, which runs from $2n$ down to $n+1$. Since $\binom{2n}{r} = \binom{2n}{2n-r}$, it becomes $\sum_{s=n+1}^{2n} s\binom{2n}{s}$, the same as the first sum. (This is the heads/tails symmetry: having s tails is as likely as having s heads.)

> **Ask yourself** How do we evaluate $\sum_{r=n+1}^{2n} r\binom{2n}{r}$, which only runs over half the range?

This is what part (i) is for. Its full-range version is given: with $2n$ in place of n, $\sum_{r=0}^{2n} r\binom{2n}{r} = 2n \cdot 2^{2n-1}$. So by (i),

$$2\sum_{r=n+1}^{2n} r\binom{2n}{r} = 2n \cdot 2^{2n-1} = n\,2^{2n}.$$

Therefore

$$2^{2n}\,\mathrm{E}(X) = n\,2^{2n} + n\binom{2n}{n}, \qquad \text{so} \qquad \mathrm{E}(X) = n\left(1 + \frac{1}{2^{2n}}\binom{2n}{n}\right),$$

as required.

Part (iii)

Let $a_n = \binom{2n}{n}/2^{2n}$, which is positive. Then

$$\frac{a_{n+1}}{a_n} = \frac{(2n+2)!}{(n+1)!\,(n+1)!} \cdot \frac{n!\,n!}{(2n)!} \cdot \frac{1}{4} = \frac{(2n+2)(2n+1)}{4(n+1)^2} = \frac{2n+1}{2n+2},$$

using $2n+2 = 2(n+1)$. This ratio is less than 1, and $a_n > 0$, so $a_{n+1} < a_n$ for every $n \geq 1$. The sequence is strictly decreasing.

Part (iv)

Your winnings are exactly the X of part (ii), and you pay £n, so the expected winnings per pound paid are

$$\frac{\mathrm{E}(X)}{n} = 1 + \frac{1}{2^{2n}}\binom{2n}{n} = 1 + a_n.$$

By (iii), a_n is largest when n is smallest, so you should choose $n = 1$. Then you get back $1 + \frac{1}{2} = \frac{3}{2}$ pounds per pound paid, an expected profit of 50p per pound. (If "winnings" means profit, $\mathrm{E}(X)/n - 1 = a_n$, and the conclusion is the same.)

Checks

- For $n = 1$ (two tosses), $X = 2$ for HH or TT and $X = 1$ for HT or TH, so $\mathrm{E}(X) = \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 1 = \frac{3}{2}$. The formula gives $1 \cdot (1 + \frac{2}{4}) = \frac{3}{2}$.
- For $n = 2$, the formula gives $2(1 + \frac{6}{16}) = \frac{11}{4}$, so the return per pound is $\frac{11}{8} = 1.375 < 1.5$. For $n = 3$ it is $\frac{21}{16} = 1.3125$, still decreasing as (iii) predicts.
- The formula for $\mathrm{E}(X)$ was confirmed for $n \le 5$ by a computer check over all 2^{2n} sequences of tosses.

Common slips

- Counting the $H = n$ term twice, or not at all. It sits on neither side of the split and contributes $n\binom{2n}{n}$ once.
- In (iv), maximising $\mathrm{E}(X)$ itself, which grows without limit, rather than $\mathrm{E}(X)$ per pound paid.

Beyond STEP

The excess $\mathrm{E}(X) - n = na_n$ is the expected distance of a simple random walk from its start, halved. By Stirling's formula, $a_n \approx 1/\sqrt{\pi n}$, so this excess grows like $\sqrt{n/\pi}$: the typical imbalance between heads and tails grows like the square root of the number of tosses. Meanwhile the excess per pound, a_n, tends to 0, which is the law of large numbers at work. Grimmett and Welsh, *Probability: An Introduction*, covers random walks and these limits.

WORKED SOLUTION

STEP III 2024, Question 12

Probability and Statistics • Geometric Probability

STEP III 2024, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

For a point chosen uniformly in the unit square, $\mathrm{P}(R \leq r)$ is just the area of the part of the square inside the circle of radius r about the origin. Once $r > 1$ the circle pokes out of two sides, and the region left inside the square splits neatly into two right-angled triangles and a sector. For the expectation, differentiate to get the density and integrate $rf(r)$; an integration by parts makes the awkward $\cos^{-1}$ term collapse, and a hyperbolic substitution finishes the job.

Solution

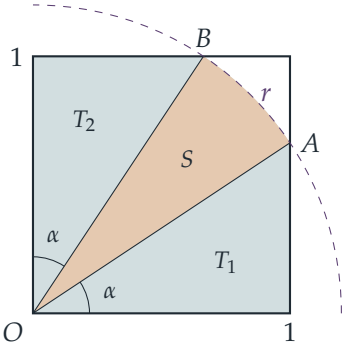
Part (i)

Since probability equals area, $\mathrm{P}(R \leq r)$ is the area of the set of points of the square at distance at most r from O.

When $0 \leq r \leq 1$ the quarter-disc $x, y \geq 0$, $x^2 + y^2 \leq r^2$ lies entirely inside the square, so

$$\mathrm{P}(R \leq r) = \tfrac{1}{4}\pi r^2 \qquad (0 \leq r \leq 1).$$

When $1 \leq r \leq \sqrt{2}$ the circle $x^2 + y^2 = r^2$ meets the side $x = 1$ at $A = \left(1, \sqrt{r^2 - 1}\right)$ and the side $y = 1$ at $B = \left(\sqrt{r^2 - 1}, 1\right)$; both lie on the square because $0 \leq \sqrt{r^2 - 1} \leq 1$. Let α be the angle OA makes with the x-axis, so $\cos\alpha = 1/r$, i.e. $\alpha = \cos^{-1}(r^{-1})$. By symmetry in $y = x$, OB makes the same angle α with the y-axis.



The part of the square inside the circle is made up of:

- triangle T_1 with vertices O, $(1,0)$, A: right-angled at $(1,0)$, with legs 1 and $\sqrt{r^2-1}$, so area $\frac{1}{2}\sqrt{r^2-1}$;
- triangle T_2 with vertices O, $(0,1)$, B: the mirror image of T_1, also of area $\frac{1}{2}\sqrt{r^2-1}$;
- the sector S of radius r between OA and OB, whose angle is $\frac{\pi}{2}-2\alpha$, so its area is $\frac{1}{2}r^2\left(\frac{\pi}{2}-2\alpha\right) = \frac{1}{4}\pi r^2 - r^2\alpha$.

Adding,

$$\mathrm{P}(R \le r) = \sqrt{r^2-1} + \tfrac{1}{4}\pi r^2 - r^2\cos^{-1}(r^{-1}) \qquad (1 \le r \le \sqrt{2}),$$

as required. (At $r=1$ both formulas give $\frac{1}{4}\pi$, and at $r=\sqrt{2}$ we get $1+\frac{1}{2}\pi - 2\cdot\frac{\pi}{4} = 1$, as we should.)

Part (ii)

Write $F(r) = \mathrm{P}(R \le r)$ and $f = F'$ for the density. On $0 < r < 1$, $f(r) = \frac{1}{2}\pi r$. On $1 < r < \sqrt{2}$ we need

$$\frac{\mathrm{d}}{\mathrm{d}r}\cos^{-1}(r^{-1}) = -\frac{1}{\sqrt{1-r^{-2}}}\cdot\left(-\frac{1}{r^2}\right) = \frac{1}{r\sqrt{r^2-1}},$$

using $\sqrt{1-r^{-2}} = \sqrt{r^2-1}/r$ for $r>0$. Then, by the product rule,

$$\begin{aligned} f(r) &= \frac{r}{\sqrt{r^2-1}} + \tfrac{1}{2}\pi r - \left(2r\cos^{-1}(r^{-1}) + \frac{r^2}{r\sqrt{r^2-1}}\right) \\ &= \tfrac{1}{2}\pi r - 2r\cos^{-1}(r^{-1}). \end{aligned}$$

The two $\dfrac{r}{\sqrt{r^2-1}}$ terms cancel, which is a good sign. Hence

$$\begin{aligned} \mathrm{E}(R) &= \int_0^{\sqrt{2}} r f(r)\,\mathrm{d}r = \int_0^{\sqrt{2}} \tfrac{1}{2}\pi r^2\,\mathrm{d}r - 2\int_1^{\sqrt{2}} r^2\cos^{-1}(r^{-1})\,\mathrm{d}r \\ &= \frac{\pi(\sqrt{2})^3}{6} - 2I = \frac{\sqrt{2}\,\pi}{3} - 2I, \end{aligned}$$

where $I = \displaystyle\int_1^{\sqrt{2}} r^2 \cos^{-1}(r^{-1})\,\mathrm{d}r$.

Ask yourself How do we get rid of the inverse cosine inside I?

Integrate by parts, differentiating $\cos^{-1}(r^{-1})$ (whose derivative we found above is algebraic) and integrating r^2:

$$\begin{aligned} I &= \left[\tfrac{1}{3}r^3 \cos^{-1}(r^{-1})\right]_1^{\sqrt{2}} - \int_1^{\sqrt{2}} \frac{r^3}{3} \cdot \frac{1}{r\sqrt{r^2-1}}\,\mathrm{d}r \\ &= \tfrac{1}{3}\cdot 2\sqrt{2}\cdot\frac{\pi}{4} - 0 - \frac{1}{3}\int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r, \end{aligned}$$

using $\cos^{-1}(1/\sqrt{2}) = \frac{\pi}{4}$ and $\cos^{-1} 1 = 0$. (The integrand is unbounded at $r = 1$, but it behaves like $1/\sqrt{2(r-1)}$ there, so the integral converges.) Therefore

$$\mathrm{E}(R) = \frac{\sqrt{2}\,\pi}{3} - \frac{\sqrt{2}\,\pi}{3} + \frac{2}{3}\int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r = \frac{2}{3}\int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r,$$

which is the required result.

Part (iii)

The $\sqrt{r^2-1}$ suggests $r = \cosh t$, with $\mathrm{d}r = \sinh t\,\mathrm{d}t$ and $\sqrt{r^2-1} = \sinh t$ (as $t \geq 0$). The limits become $t = 0$ and $t = T$, where $\cosh T = \sqrt{2}$, so $\sinh T = 1$ and

$$T = \cosh^{-1}\sqrt{2} = \ln\bigl(\sqrt{2} + \sqrt{2-1}\bigr) = \ln\bigl(\sqrt{2}+1\bigr).$$

Then

$$\begin{aligned} \int_1^{\sqrt{2}} \frac{r^2}{\sqrt{r^2-1}}\,\mathrm{d}r &= \int_0^T \frac{\cosh^2 t}{\sinh t}\,\sinh t\,\mathrm{d}t = \int_0^T \tfrac{1}{2}(1+\cosh 2t)\,\mathrm{d}t \\ &= \left[\tfrac{1}{2}t + \tfrac{1}{2}\sinh t\cosh t\right]_0^T = \tfrac{1}{2}\ln\bigl(\sqrt{2}+1\bigr) + \tfrac{1}{2}\cdot 1\cdot\sqrt{2}. \end{aligned}$$

Multiplying by $\frac{2}{3}$,

$$\mathrm{E}(R) = \tfrac{1}{3}\Bigl(\sqrt{2} + \ln\bigl(\sqrt{2}+1\bigr)\Bigr),$$

as required.

Checks

- F is continuous at $r = 1$ (both pieces give $\frac{\pi}{4} \approx 0.785$) and reaches 1 at $r = \sqrt{2}$, the far corner.
- The density $\frac{1}{2}\pi r - 2r\cos^{-1}(r^{-1})$ is zero at $r = \sqrt{2}$, as it must be: only the single corner point is that far away.

- Numerically, $\mathrm{E}(R) = \frac{1}{3}(1.41421 + 0.88137) \approx 0.7652$, sensibly between $\frac{1}{2}$ and $\sqrt{2}$. A computer simulation of 4×10^5 random points gave a mean distance of about 0.766, and its empirical distribution function agreed with F to within 0.002 at several values of r.

Common slips

- Taking the sector angle to be $\frac{\pi}{2} - \alpha$ instead of $\frac{\pi}{2} - 2\alpha$: there is a triangle cut off at *each* of the two sides.
- Getting the sign of the derivative of $\cos^{-1}(r^{-1})$ wrong: the two minus signs (from $\cos^{-1}$ and from r^{-1}) cancel, so it is positive.

Beyond STEP

This is a first taste of geometric probability: the mean distance from a corner of the unit square to a random point inside it. The companion problem, the mean distance between *two* random points in the unit square, has a similar but more elaborate closed form involving $\ln(1 + \sqrt{2})$, and it is a classic exercise in double integrals and changes of variable. Grimmett and Welsh, *Probability: An Introduction*, covers distributions defined by geometry and how to compute expectations from distribution functions.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

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